Master of Science in Computer Science
October 2026



AI-Enhanced Sleeping Cell Detection in Zero-Traffic Scenarios for Advanced Cell Supervision Systems

Hemanth Kumar Anne

Faculty of Computing, Blekinge Institute of Technology, 371 79 Karlskrona, Sweden

This thesis is submitted to the Faculty of Computing at Blekinge Institute of Technology in partial fulfillment of the requirements for the degree of Master of Science in Computer Science. The thesis is equivalent to 20 weeks of full-time studies.

The author declares that he is the sole author of this thesis and that he has not used any sources other than those listed in the bibliography and identified as references. He further declares that he has not submitted this thesis at any other institution to obtain a degree.

Contact Information:
Author(s):
Hemanth Kumar Anne
E-mail: heae24@student.bth.se

University advisor:
Associate Professor Sadi Alawadi
Department of Computer Science

Faculty of Computing
Blekinge Institute of Technology
SE–371 79 Karlskrona, Sweden

Internet : www.bth.se
Phone : +46 455 38 50 00
Fax : +46 455 38 50 57

Abstract

Background. Automated supervision infers cell health from the KPI counters each cell reports. A sleeping cell continues to appear operational while it fails to serve users, and raises no alarm of its own. Such a cell is identified from what its counters record, which requires the cell to be carrying traffic. During periods of no traffic the counters register nothing, whether the cell has failed or simply has no users to serve, leaving the two conditions indistinguishable to supervision.
Objectives. To establish whether the counter data available to supervision differentiates the two conditions, to design a method that identifies sleeping cells during zero traffic within a bounded detection time, and to evaluate its behaviour and the conditions under which it applies.
Methods. The work follows Design Science Research Methodology on counter records from an operational Long Term Evolution (LTE) network. The interval preceding zero-traffic onsets was examined across 37,585 windows from 741 cells using two unsupervised detectors resting on different principles. A detection method was then designed that fits a forecasting model to each cell's own history and assesses an observed zero against the level of activity that cell would ordinarily reach at that moment. It was evaluated through exhaustive fault injection over a seven-day window and through a run on the same week left unmodified.
Results. In the interval preceding a zero-traffic onset, neither detector found a group of windows standing apart clearly enough to classify an individual window, so no basis was found for separating a failed cell from an idle one. The method requires a forecast meeting two conditions, satisfied for twenty-nine of the fifty cells examined. Every one of the 19,488 injected faults was reported, either at the moment traffic stopped or at a deadline derived from the cell's own forecast, with no delay exceeding one daily cycle. This follows from the construction of the rule. Faults that began when the cell was expected to be busy were reported immediately, and the rest at a deadline. For faults reported at a deadline, the mean delay ranged from 3 hours 42 minutes to 10 hours 50 minutes across cells. On the unmodified week, three alarms arose on one cell across 104 zero-traffic episodes while the other twenty-eight produced none, and the episodes closing without an alarm did so with a median margin of five hours.
Conclusions. The interval before a zero-traffic onset did not indicate whether the cell had failed, so the distinction has to be drawn from elsewhere. The two cases are separated by the timing of the zero relative to the cell's own established pattern. A forecast fitted to that pattern makes the distinction operational, deciding whether a zero is anomalous at once and otherwise how long to wait. The detection outcomes were obtained on twenty-nine cells over one week, with faults introduced at known times rather than observed.

Keywords: sleeping cells, cell outage detection, anomaly detection, time-series forecasting, self-organising networks

Data Availability Statement

The data used in this study consists of operational KPI counter records from a commercial LTE network, provided by Ericsson for this thesis. It is proprietary and subject to confidentiality obligations under the terms of the industrial collaboration, and cannot be made publicly available. Site identifiers, cell identifiers, the geographical location of the network and the calendar period covered by the observations are withheld throughout for the same reason.

The source code was developed on infrastructure provided by the industrial partner and is likewise not publicly available under the same non-disclosure agreement.

The counters used, the reporting granularity, the processing applied to them, and the parameters of every model fitted are described in Chapter 4 in sufficient detail for the method to be reproduced on comparable data.

Generative AI Disclosure

The methodological design, the implementation of the analysis and the interpretation of the results are the work of the author. Generative AI tools were used in a supporting role, and example prompts are given in Appendix C.

Anthropic's Claude Opus 5 model was used for grammar correction and language polishing, and to create the detection logic figure, which the author then revised. The model was sometimes used to correct tense errors and to rephrase parts of the text to bring them closer to academic conventions in English, but was never used to generate entire sections from scratch. It was also used to learn new technical concepts. Finally, it was used to assist with LaTeX markup editing.

NotebookLM was used to fetch information from the literature. Its responses were checked directly against the source documents that NotebookLM referenced.

Grammarly was used to correct grammar and improve sentence structure. Mermaid AI was used to create the pipeline diagram, which the author then revised.

Acknowledgments

I would like to extend my heartfelt gratitude to everyone who made this thesis possible.

My sincere thanks go to my manager and supervisor at Ericsson, Robert Juretic, for his constant support throughout this work and for the trust he placed in me to carry it out.

I am grateful to my supervisors at Ericsson, Andreas Nyberg and Weaam Bayaa, for the many discussions that helped me explore ideas as the work developed, and for sharing their knowledge of how the network behaves in practice. I appreciate their support in refining ideas and testing them with various operational teams.

I would also like to thank my university supervisor, Associate Professor Sadi Alawadi, for his feedback, and for his guidance and support whenever I needed it throughout the project.

Finally, my deepest thanks go to my mother, father and brother, and to my friends, for their patience and encouragement.

Contents

Abstract i

Data Availability Statement iii

Generative AI Disclosure v

Acknowledgments vii

List of Acronyms xvii

1 Introduction 1
- 1.1 Problem Statement . . . 1
- 1.2 Aim and Objectives . . . 2
  - 1.2.1 Aim . . . 2
  - 1.2.2 Objectives . . . 2
- 1.3 Research Questions . . . 2
- 1.4 Ethical, Societal, and Sustainability Aspects . . . 3
  - 1.4.1 Ethical Considerations . . . 3
  - 1.4.2 Societal Aspects . . . 3
  - 1.4.3 Sustainability Impact . . . 4
- 1.5 Scope and Delimitations . . . 4
- 1.6 Outline . . . 5

2 Background 7
- 2.1 Self-Organising Networks . . . 7
- 2.2 Cellular Network Supervision . . . 8
- 2.3 Sleeping Cells and Zero-Traffic Ambiguity . . . 9
- 2.4 Unsupervised Anomaly Detection . . . 9
  - 2.4.1 Isolation Forest . . . 10
  - 2.4.2 Local Outlier Factor . . . 10
- 2.5 Time-Series Forecasting . . . 11
  - 2.5.1 Time-Series Structure and Decomposition . . . 11
  - 2.5.2 Forecasting and Model Requirements . . . 12
  - 2.5.3 Forecasting Methods . . . 13
  - 2.5.4 Forecast Accuracy . . . 14

3 Related Work **17**
3.1 Sleeping Cell Detection Methods . . . 17
3.1.1 Detection from Device Measurements . . . 17
3.1.2 Detection from Neighbouring Cells . . . 19
3.1.3 Detection from Reported KPIs . . . 19
3.2 Research Gap and Thesis Positioning . . . 22

4 Method **25**
4.1 Research Design . . . 25
4.1.1 Design Science Research Methodology . . . 25
4.1.2 Mapping Research Questions to Study Activities . . . 26
4.1.3 Alternative Methodologies . . . 26
4.1.4 Literature Survey Protocol . . . 27
4.2 Problem Identification (Activity 1) . . . 27
4.2.1 Advanced Cell Supervision . . . 27
4.2.2 Supervision Data . . . 28
4.2.3 Zero-Traffic Conditions . . . 29
4.2.4 Absence of Fault Labels . . . 30
4.3 Exploratory Analysis of Pre-Zero Behaviour (Activity 1) . . . 30
4.3.1 Window Extraction . . . 30
4.3.2 Normalisation . . . 31
4.3.3 Feature Construction . . . 31
4.3.4 Anomaly Detection . . . 32
4.3.5 Comparison Procedure . . . 33
4.4 Detection Method Design and Implementation (Activity 3) . . . 33
4.4.1 Regularity in Cell Traffic . . . 34
4.4.2 Detection Principle . . . 36
4.4.3 Detection Logic . . . 37
4.4.4 Forecast Model Selection . . . 39
4.4.5 Model Configuration and Applicability . . . 42
4.4.6 Pipeline Implementation . . . 44
4.5 Demonstration and Evaluation (Activities 4 and 5) . . . 46
4.5.1 Evaluation Strategy . . . 46
4.5.2 Detection with Injected Faults . . . 47
4.5.3 Detection on Unmodified Data . . . 47

5 Results and Analysis **49**
5.1 Behaviour Preceding a Zero-Traffic Onset . . . 49
5.2 Applicability . . . 50
5.3 Detection with Injected Faults . . . 51
5.4 Detection on Unmodified Data . . . 51
5.5 Margin on Resolved Episodes . . . 52

6 Discussion **55**
6.1 Answers to the Research Questions . . . 55
6.2 Interpretation of the Detection Outcomes . . . 57
6.3 Relation to Existing Work . . . 57

6.4 Threats to Validity 59
6.5 Limitations 60
6.6 Ethical, Societal, and Sustainability Aspects 61

7 Conclusions and Future Work **63**
7.1 Contributions 63
7.2 Practical Implications 64
7.3 Future Work 64

References **67**

A Supplemental Information **71**
A.1 Pre-Zero-Traffic Feature List 71
A.2 Feature Rankings 72
A.3 Model Comparison Breakdown 74

B Per-Cell Injection Results **75**

C Prompt Examples **77**
C.1 Example: Rephrasing for Academic Tone 77
C.2 Example: Proofreading for Formatting and Punctuation 78

List of Figures

4.1 The six activities of the DSRM as applied in this study, adapted from Peffers et al. [33]. 26
4.2 Recorded RRC connection setup attempts over the full observation period of 45 days, for (a) a cell with median non-zero value 536 and (b) a cell with 1754. 34
4.3 The same two cells over a single week. 35
4.4 Multiple seasonal-trend decomposition for the first cell, showing the observed series, trend, daily component, weekly component, and remainder. 35
4.5 Multiple seasonal-trend decomposition for the second cell. 36
4.6 Assessment of two zero-traffic onsets across a daily cycle, shown schematically. The horizontal segments at zero show the observed indicator for each onset. Onset A falls where the forecast exceeds the threshold and raises an alarm at once. Onset B falls below it, and the deadline is set by the first crossing. 39
4.7 Arrangement of the detection system. The model store is written once for each cell from its recorded history and read thereafter, so no model fitting takes place while reports are being assessed. Cells routed to the fallback path bypass forecast generation. 44

5.1 Cell M15 over the period containing the three alarms. The observed series and the forecast are shown against the cell's activity threshold (shown as the median), with the three false alarms (FA) marked. . . . 52
5.2 The episode with the smallest margin, on cell M5. The observed series and the forecast are shown against the cell's activity threshold, with the onset, the resumption and the deadline marked. 53

List of Tables

3.1 Overview of reviewed sleeping cell detection work 22

4.1 Research questions mapped to the activities addressing them 26
4.2 Distribution of cells across regions. 28
4.3 KPI counters available for each cell. 29
4.4 Measures computed for each counter over the window. 31
4.5 Seasonal strength across the fifty cells 36
4.6 Detection outcomes by condition at the onset 38
4.7 Comparison protocol 41
4.8 MASE across the fifty cells 41
4.9 Parameters varied and values examined 42
4.10 Prophet at default and tuned configuration across the fifty cells . . . 43
4.11 Situations in which a zero is observed and how each resolves 45
4.12 Division of the observation period 47

5.1 Routing outcome across the fifty cells 50
5.2 The four cells failing the suitability condition. The gap is the shortfall of the forecast maximum against the threshold, as a percentage of the threshold. 50
5.3 Aggregate outcome of the 19,488 injection runs across the twenty-nine cells 51
5.4 The three onsets that raised an alarm on the unmodified week, all on cell M15. Onset times correspond to those marked in Figure 5.1. . . . 52
5.5 Distribution of margin across the 101 episodes that closed without an alarm. Margins are counted in 15-minute reporting intervals. 53

A.1 Counter prefixes used in feature identifiers 71
A.2 Measure suffixes used in feature identifiers 71
A.3 Cross-signal and divergence features 72
A.4 Feature rankings for Isolation Forest at the 10% threshold 72
A.5 Feature rankings for Local Outlier Factor at the 10% threshold 73
A.6 Cells reached, by combination of candidates 74

B.1 Per-cell breakdown of the injection runs across the 29 modellable cells. Immediate is the share of injection runs on the cell whose onset was on the immediate path. Deferred is the complement. 75

List of Acronyms

3GPP	3rd Generation Partnership Project
ACS	Advanced Cell Supervision
AI	Artificial Intelligence
ARIMA	Autoregressive Integrated Moving Average
ARMA	Autoregressive Moving Average
CBRA	Contention-Based Random Access
DL	Downlink
DSRM	Design Science Research Methodology
eNodeB	Evolved NodeB
ETS	Error, Trend, Seasonal
GSM	Global System for Mobile Communications
KPI	Key Performance Indicator
LOF	Local Outlier Factor
LTE	Long Term Evolution
MAE	Mean Absolute Error
MAPE	Mean Absolute Percentage Error
MASE	Mean Absolute Scaled Error
MB	Megabyte
MDT	Minimization of Drive Tests
MSTL	Multiple Seasonal-Trend Decomposition using Loess
NDA	Non-Disclosure Agreement
NR	New Radio
OSS	Operations Support System

QoE	Quality of Experience
RACH	Random Access Channel
RAN	Radio Access Network
RMSE	Root Mean Squared Error
ROP	Result Output Period
RQ	Research Question
RRC	Radio Resource Control
RSRP	Reference Signal Received Power
RSRQ	Reference Signal Received Quality
SINR	Signal to Interference plus Noise Ratio
SON	Self-Organising Network
STL	Seasonal-Trend Decomposition using Loess
TBATS	Trigonometric Seasonality, Box-Cox Transformation, ARMA Errors, Trend and Seasonal Components
UE	User Equipment
UMTS	Universal Mobile Telecommunications System

Chapter 1

Introduction

Modern cellular networks are operated at a large scale and continue to grow in size and structural complexity. Thousands of cells are deployed across urban, suburban and rural environments [41, 42]. Each of these cells must be monitored so that faults are caught before affecting users. However, traditional network supervision is done manually by analysing alarm codes and establishing static thresholds, which is increasingly unsustainable and error-prone [9, 41, 44]. A fault found through manual supervision is usually found late, after the service loss has already been affecting some users [7]. Managing these deployments manually is no longer technically or financially sustainable for operators, largely because of the massive increase in network size and the diverse mix of cell technologies [2]. Modern mobile network operators have been moving from human-reliant monitoring towards automated supervision and AI-supported network health monitoring systems [32].

Automated supervision systems observe heterogeneous streams of spatiotemporal data that mobile networks naturally generate during operation [42], such as performance counters, and examine them continuously to identify cells that are not behaving as expected [13]. Many current supervision systems integrate machine learning to dynamically learn normal behaviour, allowing them to separate faults [32]. The shift toward proactive, AI-driven anomaly detection is particularly important [5] because not all network degradation events trigger explicit hardware or software alarms [2].

One important class of faults in radio access networks is the sleeping cell. A sleeping cell is a cell that appears technically operational to monitoring systems but fails to serve users due to underlying issues such as hardware degradation, configuration errors, or software defects [10, 31]. These faults are particularly challenging to detect because the sleeping cells generate no explicit alarms while causing service degradation. They often remain invisible to standard operation and maintenance entities until user complaints arise. When undetected, these silent failures can persist for extended periods. This leads to severe degradation of Quality of Experience (QoE) and increased load on neighbouring cells that must absorb the displaced traffic [28]. To overcome this lack of direct visibility, automated supervision systems leverage indirect patterns in network data, such as deviations in Key Performance Indicators (KPIs) [9].

1.1 Problem Statement

The current Advanced Cell Supervision (ACS) system at Ericsson detects sleeping cells effectively when user traffic is present, but cannot reliably distinguish a healthy

idle cell from a sleeping cell during zero-traffic periods. The core difficulty is that, at the moment a zero-traffic period begins, no observation of the cell-side traffic signal can separate the two conditions, because the recorded data is identical in both cases. A healthy idle cell records zero because no one is trying to use it, and a sleeping cell records zero because it cannot serve the users that are present. This creates a supervision blind spot where both conditions appear identical to the system, leading to potential false alarms on legitimately idle cells or missed detection of actual faults. Detection during zero traffic must therefore rely on information beyond the zero observation itself. The problem addressed in this thesis is the development of a detection method that identifies sleeping-cell behaviour during zero-traffic periods while limiting the alarms raised on cells that are legitimately idle.

1.2 Aim and Objectives

1.2.1 Aim

The aim of this thesis is to enhance Advanced Cell Supervision by developing a detection method that uses each cell's own historical traffic pattern to judge whether an observed zero is expected or anomalous, thereby distinguishing healthy idle cells from sleeping cells during zero-traffic scenarios.

1.2.2 Objectives

To achieve the aim, the thesis pursues the following objectives:

- Analyse zero-traffic behaviour in operational cellular network data and characterise the conditions under which current supervision mechanisms cannot distinguish healthy idle cells from sleeping cells.
- Investigate whether features or patterns extracted from the available network data provide discriminative power to differentiate healthy zero-traffic from fault-related zero-activity.
- Design and implement a detection approach that identifies sleeping-cell conditions during zero-traffic scenarios within a bounded detection time.
- Evaluate the detection behaviour of the proposed approach on zero-traffic scenarios, including the alarms raised on cells that are legitimately idle and the conditions under which the approach applies.

1.3 Research Questions

The aim is explored through the following research questions:

- **RQ1.** What AI-based approach can distinguish sleeping cells from healthy idle cells during zero-traffic periods in cellular networks?

Justification. A zero-traffic observation carries no information that separates a healthy idle cell from a sleeping cell, since neither condition produces activity for supervision systems to examine. This question asks what a detection method can rely on when the zero itself reveals nothing.

- **RQ2.** What features or patterns extracted from cellular network data can effectively differentiate healthy zero-traffic cases from fault-related zero-activity conditions?

  Justification. Standard KPI counters read zero under both conditions during zero-traffic periods, so the reported values alone do not separate them. Whether anything else in the data carries the distinction is therefore an open question, and one that must be settled before a detection method can be designed.

- **RQ3.** What detection performance does the proposed approach achieve in distinguishing healthy idle cells from sleeping cells during zero-traffic periods?

  Justification. Sleeping cell detection in zero traffic is time-sensitive, since the longer a fault persists undetected, the greater the service impact on the users a cell serves. A proposed method therefore requires empirical evidence of how quickly it reports a fault and how it behaves on cells that are legitimately idle.

1.4 Ethical, Societal, and Sustainability Aspects

1.4.1 Ethical Considerations

This research is conducted under a signed non-disclosure agreement (NDA) with Ericsson AB and follows BTH's research ethics guidelines throughout.

Confidentiality. The operational network data used in this work is the property of Ericsson AB and was accessed and processed on infrastructure provided by the industrial partner. Configuration and deployment details that could identify specific network infrastructure are withheld in accordance with the NDA, and all results are reported in a form suitable for academic publication.

Anonymisation of network data. The dataset identifies cells and sites by their operational identifiers. These identifiers, along with site names, geographic areas and the operator's identity, are anonymised or replaced with generic labels wherever they would appear. The data itself consists of aggregate KPI counters recorded at cell level and contains no subscriber or device identifiers, so no information relating to an identifiable individual is processed at any stage. The calendar period covered by the observations is also withheld under the NDA.

1.4.2 Societal Aspects

Reliability. Automated supervision exists because faults that remain unnoticed translate into people losing service. The time between a fault occurring and the operator becoming aware of it determines how many users are affected and for how long. For sleeping cells, this time depends on how quickly the supervision system

recognises the fault. Extending supervision to zero-traffic periods reduces how long a fault of this kind can persist unnoticed.

Access to emergency services. A sleeping cell cannot carry emergency calls from its coverage area. Zero-traffic periods are also the periods during which few users are present to report a problem, so the likelihood of the fault being reported by an affected user is lowest at exactly the time when supervision cannot observe it.

1.4.3 Sustainability Impact

Cost of late detection. A sleeping cell that supervision does not detect is handled reactively, after user complaints or through manual inspection [29]. Since the fault raises no alarm, resolving it depends on human diagnostic effort, and this effort grows the longer the fault remains unnoticed [2]. The cause cannot be established remotely, inspection at the site involves travel [35], and the energy and emissions this carries are part of what late detection costs in terms of overall network sustainability.

Inefficiency of undetected outages. When a cell fails silently, it remains powered but the traffic it should have carried is displaced onto neighbouring cells, which operate at higher load to absorb it [28]. The vast majority of operational power is consumed directly at base station sites, representing the dominant driver of the network's overall energy footprint [4]. A cell that serves no users therefore continues to consume close to what it would consume in service, and this persists for as long as the fault goes unnoticed.

1.5 Scope and Delimitations

The following boundaries define what this work covers.

Detection condition. The work addresses zero-traffic periods only, a condition where cell-side KPI counters record zero and the existing ACS system cannot distinguish a healthy idle cell from a silently failed one. Detection while a cell is carrying traffic is handled by existing ACS and is not revisited here. The contribution extends supervision into a condition it does not currently cover rather than replacing what already operates.

Fault model. The work targets faults in which service ceases and does not resume. Intermittent faults, partial service degradation, and faults that do not result in complete traffic loss are outside scope.

Network data. The work is carried out on operational 4G LTE data provided by Ericsson, originating from a single operator. Detection in 5G New Radio (NR) or other network generations is not addressed. Transferability to networks with different deployment characteristics or reporting configurations is not established here.

Observable signals. Detection is based on cell-level KPI counters already collected during network operation at 15-minute reporting intervals. Signals requiring additional infrastructure, configuration changes, or dedicated measurement campaigns are not considered. The system operates on what is already available without imposing new data collection requirements.

Supervision stage. The work concerns detection alone. Root cause analysis and recovery actions are separate stages of supervision and are not attempted.

Absence of labels. The operational data carries no labels identifying past sleeping cell occurrences. The work is therefore delimited to methods that can be developed and evaluated without labelled cases.

1.6 Outline

The remainder of this thesis is structured as follows. Chapter 2 introduces the background concepts needed to follow the thesis. Chapter 3 reviews existing work on sleeping cell detection and positions the zero-traffic gap within that literature. Chapter 4 describes the research methodology adopted and presents the design and development of the proposed detection approach. Chapter 5 presents the outcome of the analyses and the evaluation of the proposed approach. Chapter 6 discusses these results in relation to the research questions and to the existing literature, and sets out the threats to validity and limitations of the work. Chapter 7 concludes with the contributions of the thesis, its practical implications, and directions for future work.

Chapter 2

Background

At its most fundamental level, a mobile telecommunications network is built on a cellular architecture. To provide wireless service across a region, the geographic area is partitioned into smaller, localised coverage zones called cells [41]. This cellular division allows the network to distribute coverage boundaries and manage radio resources based on the local user population density and physical radio propagation limits [41].

In 4G Long Term Evolution (LTE) networks, a cell is served by a base station termed an Evolved NodeB (eNodeB). The eNodeB allocates radio resources to the devices operating within its coverage and carries their data traffic over the air interface [41]. The area a cell covers is determined by the antenna serving it. An omnidirectional antenna transmits across a full 360-degree field, so that a site fitted with one serves a single cell. Modern sites more commonly carry directional antennas, each covering a sector of roughly 120 degrees, so that one site is divided into three sectors and each is operated as a cell in its own right [9, 27, 29]. Because a sector has its own antenna, a fault may affect one while the other cells on the same site continue to operate [29]. A cell is therefore the fundamental logical unit for delivering service within a region and for tracking network operations.

To establish a connection to the Radio Access Network (RAN), a User Equipment (UE), the subscriber's mobile device, must set up an active radio connection with the serving eNodeB before transmitting and receiving data [41]. This attachment process is initiated by the UE scanning its surrounding environment and selecting the cell sector with the highest measured Reference Signal Received Power (RSRP) [3, 9].

As demand for mobile data exploded, operators have deployed more base stations and cells to maintain coverage and capacity. Managing, configuring and maintaining a dense infrastructure at this scale entirely by hand is no longer feasible. Manual troubleshooting is financially unsustainable, consuming a large share of the operational budget. It is also too slow to catch and resolve the many localised equipment, configuration and software faults that can degrade service across a region [2, 29].

2.1 Self-Organising Networks

To address the complexity of manual operation and maintenance, the telecommunications industry has introduced the Self-Organising Networks (SON) paradigm. The SON architecture is designed to minimise human intervention and is fundamentally built upon three functions [2, 15].

Self-Configuration.
Supports the deployment of new network elements by automating the initial setup tasks required before operation. These tasks include establishing network connectivity, obtaining configuration data, and preparing radio parameters so that the network element can be integrated into the existing network with limited manual effort [15].

Self-Optimisation.
Focuses on improving network operation by continuously monitoring network conditions and adjusting configuration parameters when needed. These adjustments help make efficient use of available network resources and adapt the network to changing conditions [15].

Self-Healing.
Focuses on maintaining network performance by automatically managing cell degradations and failures. It includes mechanisms for detecting abnormal network behaviour, identifying the root causes of faults, and applying recovery or compensation actions to reduce service disruption [15].

The work presented in this thesis lies within self-healing, and specifically within its first stage, the detection of faults.

2.2 Cellular Network Supervision

Within the self-healing pillar, the continuous monitoring of the network is managed by the Operations Support System (OSS) [32]. Instead of relying solely on hardware fault alarms, this supervision operates by continuously ingesting operational data to learn normal behaviour and report cells whose current performance departs from expected norms [9, 41].

The operational telemetry originates as a continuous stream of discrete signaling and traffic events generated by individual user connections [7]. However, because transmitting this raw, event-level data in real time can severely overload the transport networks and centralised management systems [13], these events are aggregated into uniform time windows to characterise network behaviour [32]. In commercial performance management, this telemetry is typically collected over standard 15-minute granularity periods [31], although alternative aggregation windows, such as five, ten, or thirty minutes, are also employed depending on specific operational and analysis requirements [28].

A counter is a value kept by the firmware or software running on a network element such as an eNodeB. A counter records events and its value increases as each occurrence is registered [15]. At the OSS these counter values are reported directly, and are also used to compute Key Performance Indicators. Such an indicator is obtained by applying a formula, frequently a standardised one, to the values of one or several counters. It most commonly expresses the rate at which a given procedure succeeds or fails [7, 15]. For instance, a count of attempts alongside a count of successes yields the rate at which that procedure succeeds. Both are used in supervision, and the term KPI counter is used in this thesis to refer to the reported

quantities collectively, whether they are raw counts of procedures or proportions derived from them.

To access network services, a UE first performs a contention-based random access procedure and then transmits a Radio Resource Control (RRC) connection request message [36]. If the request is accepted, the eNodeB responds with a setup message containing the initial radio resource configuration, thereby establishing an RRC connection between the UE and the serving cell [36]. The volume of such requests is among the quantities supervision systems monitor [41]. In this thesis, this procedure is referred to as an RRC connection attempt, consistent with the terminology used for the corresponding KPI counters.

2.3 Sleeping Cells and Zero-Traffic Ambiguity

Sleeping cells differ in how much service survives the fault, and the literature separates three degrees. The mildest, termed an impaired cell, continues to serve users while performing below expectation, whereas a crippled cell remains in service at severely reduced capacity. In the most severe case, the catatonic cell, service ceases altogether and the cell carries no traffic [9, 10]. This thesis addresses the catatonic case. A failure of the Random Access Channel (RACH), the channel LTE networks use for initial connection, prevents new users from connecting while those already attached continue to be served. The cell, which is at first merely crippled, becomes catatonic once all the remaining connections end [9, 28].

In the counters described in Section 2.2, such a cell leaves no trace. Every counter recording activity reads zero for as long as the fault lasts. These counters record successive steps of the same access procedure, so they rise and fall together and reach zero at the same time [41].

The difficulty is that a healthy cell with no users requesting service produces the same record. Zero traffic is not in itself abnormal. Demand follows daily and weekly cycles that fall to their lowest in the early hours [42], and cells serving sparsely populated areas may see little demand at any hour [22]. In both situations the cell is idle rather than faulty.

The two conditions cannot be separated because of what a counter measures. A counter only registers a procedure the cell took part in. If no UE tries to connect, nothing is registered. If a UE tries but the cell cannot receive the attempt, nothing is registered either. Both leave zero in every activity counter available to supervision, and zero tells supervision that nothing was registered without telling it why.

2.4 Unsupervised Anomaly Detection

Faults of the kind described in Section 2.3 leave no record identifying them, and confirmed fault labels are in any case rare in operational cellular network data at the scale that performance telemetry accumulates [22, 41]. A learned decision boundary requires labelled examples of the condition it is expected to recognise, so no such boundary can be trained here. Without reliable ground-truth labels, the necessary alternative is to characterise routine network behaviour and treat significant departures from it as candidate anomalies [7, 22].

Unsupervised algorithms operate on the assumption that anomalous patterns are rare occurrences that differ significantly from the bulk of normal instances [24]. A dataset does not itself declare which of its observations are unusual. Each algorithm supplies that definition, and it depends on how the algorithm treats the feature space. Density-based methods score an observation by how sparse its own neighbourhood is relative to those of its neighbours [8], whereas isolation-based methods score it by the partitioning effort required to separate it from the remainder of the data [24]. Because the two methods rest on fundamentally different principles, an observation flagged as an anomaly under one does not necessarily stand apart under the other.

2.4.1 Isolation Forest

Traditional model-based anomaly detection methods typically rely on constructing a profile of normal data patterns and identifying any records that deviate from this baseline as anomalous. In contrast, the Isolation Forest algorithm, proposed by Liu, Ting, and Zhou, does not build a profile of normal data. It isolates anomalies from the rest of the dataset directly [24].

The algorithm constructs an ensemble of Isolation Trees to isolate anomalies. Each tree is built by recursively partitioning a sample of the dataset. At each internal node, an attribute is chosen at random, and a split value is then drawn at random between the minimum and maximum values of that attribute in the current sample. The splitting continues until each point stands alone, the remaining points are identical, or the tree reaches a predefined height limit [24].

Under this random partitioning scheme, anomalies require significantly fewer splits to be isolated, which places them at shallow depths near the root of the tree. Conversely, normal observations are clustered more densely and require many more partitions, which places them at the deeper end of the tree structure [24].

The path length of an observation is the number of edges it traverses from the root node to a terminating external node, and so records how many partitions were needed to isolate it. Because each tree is built from random splits, the path length in a single tree is unreliable. It is therefore averaged over all trees to give the expected path length, which becomes stable as more trees are added. The anomaly score is computed from this expected path length, and the shorter the path, the higher the score. Ranking observations by score is therefore the same as ranking them by how easily they are isolated [24].

The mechanism rests entirely on isolability measured against the sample as a whole. Isolation Forest employs no distance or density measure at any stage [24], so an observation is judged by how readily random partitioning separates it from the remainder of the data rather than by reference to the observations immediately surrounding it. It does not ask how that observation compares with the points nearest to it, and this second comparison is the basis of a different approach.

2.4.2 Local Outlier Factor

The Local Outlier Factor (LOF) algorithm judges each observation against its immediate surroundings rather than against the dataset as a whole, using the distances between the observation and its nearest neighbours [8].

An observation in a crowded region has near neighbours, while one in a sparse region has distant ones. LOF uses this to give each observation a local density, based on its distances to a set number of nearest neighbours. A low density alone does not make an observation unusual, since some regions of the data are naturally sparser than others. LOF therefore also computes the local density of each neighbour and compares the observation against them [8].

The comparison takes the form of a ratio, dividing the average local density of the neighbours by that of the observation itself. An observation whose region is about as dense as those of its neighbours receives a score close to one, while one whose region is noticeably sparser than theirs receives a larger score [8]. Every observation therefore receives a score, and the scores order the whole dataset by relative local density.

Although the two algorithms work differently, they produce the same kind of output. Rather than returning binary classifications, both Isolation Forest and Local Outlier Factor assign a continuous score to every observation, representing its relative degree of isolation or local density deviation. The observations are then placed in order by that score [8, 24]. To arrive at a set of candidates, a threshold is applied to this ranking, usually expressed as the proportion of the data to be treated as anomalous [24]. Since that proportion is supplied rather than discovered, the highest-ranked fraction of any dataset can be extracted regardless of whether those observations differ meaningfully from the remainder. The existence of candidates is therefore not in itself a finding, and their separability from the remaining observations must be established independently.

In summary, Isolation Forest relies on global partitioning to measure how easily an observation can be isolated from the dataset as a whole [24], whereas Local Outlier Factor compares an observation's density with that of its immediate neighbourhood [8]. Methods resting on these different principles are in practice often applied together [25].

It is important that the candidates found are meaningfully useful rather than just being different. A test of statistical significance can determine whether an observed difference is real, but such tests are inherently sensitive to sample size. Given enough observations, even a small real difference will be found significant [37]. The magnitude of that difference is what governs its utility, since a difference can be real without being large enough to be meaningful. A small separation means that the two distributions overlap heavily, so most observations fall in a region shared by both and cannot be assigned to either.

2.5 Time-Series Forecasting

2.5.1 Time-Series Structure and Decomposition

A univariate time series is a sequence of observations recorded at fixed, successive intervals [19]. Because the observations come in time order, every observation is related to those that come before it, and that is what gives the series its patterns. Seasonality describes one such pattern, repeating systematically over a fixed time interval. The number of observations required to complete one full cycle of this

pattern is the seasonal period [19].

When data is sampled at high frequencies, a single time series can exhibit multiple seasonal periods simultaneously [6, 12]. For example, if observations are recorded at fifteen-minute intervals, the data becomes fine-grained enough to carry concurrent cycles: a daily cycle spans a seasonal period of 96 observations, while a weekly cycle spans 672 observations.

To examine these underlying structures, an observed time series is often subjected to decomposition. This procedure separates the raw sequence into distinct components: a seasonal component capturing periodic fluctuations, a trend capturing the long-term direction of the mean, and a remainder containing the residual variation [11]. Each component can then be examined on its own [19].

Standard decomposition techniques such as Seasonal-Trend decomposition using Loess (STL) can handle only a single seasonality. Multiple Seasonal-Trend decomposition using Loess (MSTL) extends the procedure to series with several concurrent periods [6]. It applies STL once for each seasonal period, starting with the shortest, and removes that period's component before moving to the next. The procedure is then repeated a few times, each time re-estimating every component with the others removed, so that the components are refined [6].

Following decomposition, each periodic pattern can be quantified by its seasonal strength, a measure introduced by Wang et al. [39]. The measure first removes the trend and works on the seasonal pattern and the remainder that are left. It assesses what share of their combined variation comes from the seasonal pattern and what share comes from the remainder. A score near 1 means the seasonal pattern accounts for most of that variation, while a score near 0 means the remainder contributes more and the seasonal pattern is weak [19, 39]. The calculation can occasionally produce a negative value, which implementations set to zero, so reported scores lie between 0 and 1 [19].

2.5.2 Forecasting and Model Requirements

A forecast is an estimate of a value that has not yet been observed, produced from the observations that came before it [19]. Once the actual value is recorded, the forecast can be compared against it. The number of steps ahead that a model is required to predict is the forecasting horizon. A point forecast gives a single predicted value for each step within the horizon [19].

Forecasting methods differ fundamentally in the seasonal structures they can represent. For a series carrying more than one seasonal period, the choice of model is constrained by whether the algorithm can represent both simultaneously [12, 19]. This choice is further governed by whether a method can accommodate the large seasonal periods that arise from fine-grained sampling [12]. Assessed against these two requirements, the most established classical approaches prove unsuitable.

The autoregressive integrated moving average (ARIMA) framework is one of the two most widely used approaches to time-series forecasting [19]. It models each value from the series' own past values and past forecast errors, after differencing the series to remove its trend. These terms reach back only a few steps, so ARIMA on its own does not model seasonality. Seasonal ARIMA adds terms that look back exactly one seasonal period, but only one such period can be specified [19]. ARIMA

therefore cannot represent a daily and a weekly cycle together, and fails the first requirement [12].

The second widely used approach is exponential smoothing, which forecasts from a weighted average of past observations, giving more weight to recent ones [19]. Its standard modern form is the error, trend and seasonal (ETS) framework, in which seasonality is an explicit component. This component keeps a separate value for every step of the cycle, for example one for each fifteen-minute interval of the day. A daily cycle in such data therefore needs 96 values and a weekly cycle 672, which is too many for the model to learn reliably from the data [19]. ETS thus fails the second requirement, that a method must handle the long seasonal periods of fine-grained data. It also cannot represent more than one seasonal period at once, which is why De Livera et al. developed a separate framework [12]. The limitation applies only when ETS must model the seasonality itself. If the seasonal components are removed from the series beforehand, ETS can forecast what remains [19].

2.5.3 Forecasting Methods

Three approaches handle multiple seasonal periods simultaneously: decomposition using loess, trigonometric state-space modelling, and harmonic regression using Fourier terms [18]. These are distinct mechanisms rather than variants of a single approach. One representative of each is described below.

MSTL with exponential smoothing. The first approach uses MSTL, described in Section 2.5.1, to separate the series into its seasonal components, trend and remainder before any forecast is made. MSTL estimates each seasonal component separately, one period at a time. Forecasting then takes place in two stages [19]. First, the estimated seasonal components are subtracted from the series, and exponential smoothing is applied to the seasonally adjusted series that remains. Second, each seasonal component is carried forward by repeating its last complete cycle across the horizon, and these are added back to the exponential smoothing forecast to give the final forecast [19].

TBATS. The second approach represents each seasonal pattern with smooth sine and cosine waves within a state-space framework, in which the parts of the model are updated step by step as each observation arrives [12]. Each seasonal period is built from a wave that repeats once per period, plus faster waves, called harmonics, that repeat several times within it. A coefficient on each wave sets its height and timing, so that together the waves reproduce the peaks and troughs of the observed pattern. The coefficients are not held as fixed constants but as state variables, so they may change gradually as the series progresses. This allows the seasonal pattern to evolve. All parts of the model, namely the seasonal waves, the trend and a component for the remaining errors, are estimated together in a single fit, rather than in stages as in the first approach. The number of harmonics for each period is chosen automatically, so that sharper seasonal shapes receive more waves, while smoother ones receive fewer. The framework also offers an optional Box–Cox transformation, which stabilises the variation of the series but requires all values to be strictly positive [12].

Prophet. The third approach treats forecasting as fitting a curve over time, rather than predicting each value from the values before it as the first two approaches do [38]. Each value is expressed as the sum of a trend and one seasonal term for each period, and all of these are fitted together in a single regression. The trend is a sequence of straight line segments joined end to end, so that its slope can change over time. The points where two segments meet are called changepoints. Prophet places candidate changepoints at regular intervals across the training period and lets the data determine how much the slope changes at each, unless they are specified by hand [30, 38]. Each seasonal term is a Fourier series, built as in TBATS from a wave that repeats once per period plus faster waves that repeat several times within it. The number of waves used sets how much detail the seasonal shape can carry. Unlike in TBATS, the coefficients on these waves are fixed once fitted, so the same seasonal shape applies across the whole training period. Without its optional holiday component, the model expresses the series as [38]

$$y(t) = g(t) + s(t) + \varepsilon(t), \tag{2.1}$$

where $g(t)$ is the trend, $s(t)$ the seasonal component, and $\varepsilon(t)$ the error.

Prophet's behaviour is governed by several settings that determine how closely the fitted model follows the data. The changepoint prior scale determines how freely the trend may bend at each changepoint. A small value holds it close to a straight line, while a larger value lets it bend to follow changes in the growth rate. The seasonality prior scale limits how large the seasonal swings are allowed to become [38].

Prophet can also combine the seasonal component with the trend in two ways. In the additive form of Equation 2.1, the seasonal component adds a fixed amount to the trend. The seasonal swing is then the same size irrespective of the level of the trend. In the multiplicative form, the seasonal component is instead a proportion of the trend. The seasonal swing then grows and shrinks with the level of the trend [30]. Prophet includes daily, weekly and yearly seasonal components by default. Any of these can be switched off, and components with other periods can be added [30].

2.5.4 Forecast Accuracy

Evaluating the accuracy of a forecasting model requires quantifying the difference between predicted values and actual observations. Forecast error metrics generally fall into two categories: scale-dependent and scale-independent measures.

Scale-dependent metrics, such as the Mean Absolute Error (MAE) and the Root Mean Squared Error (RMSE), quantify the error in the original units of the data. These measures are useful for assessing the accuracy of forecasts for an individual time series. However, because they are bound to the scale of the underlying data, they cannot be meaningfully compared across multiple time series that operate at different magnitudes [20]. An error of 100 may represent a large error on a series with low average values, but a relatively small error on a series with substantially larger values.

Percentage-based metrics such as the Mean Absolute Percentage Error (MAPE) address this by expressing the error relative to the observed value. MAPE is calculated by dividing the absolute error at each time step by the actual observation

at that same time step, and then averaging these percentages over the evaluation period. While this approach allows for comparison across series of different scales, it introduces a critical mathematical limitation. Because the actual observation serves as the denominator for each individual calculation, any instance where the actual value is zero makes the corresponding error mathematically undefined [20]. Consequently, percentage-based measures such as MAPE are unsuitable for time series where zero values occur naturally.

To resolve both the scale variance and the division-by-zero limitations, the Mean Absolute Scaled Error (MASE) is employed [20]. MASE evaluates the forecast error relative to the in-sample error of a simple naive baseline model. For seasonal series, the usual baseline is the seasonal naive forecast, which predicts the value at time t as the value observed one seasonal period earlier, at time $t - m$, where m is the length of the seasonal period [19]. If a series carries more than one seasonal period, the cycle used for the baseline must itself be explicitly chosen. Assuming a historical training period of T observations and a forecast horizon of J steps, the formula for MASE is defined as [19, 20]:

$$\text{MASE} = \frac{\frac{1}{J}\sum_{j=1}^{J} |y_{T+j} - \hat{y}_{T+j}|}{\frac{1}{T-m}\sum_{t=m+1}^{T} |y_t - y_{t-m}|} \tag{2.2}$$

Here y_{T+j} and $\hat{y}_{T+j}$ denote the actual and forecast values at step j of the evaluation period. The numerator is the mean absolute error of the forecast over that period, and the denominator the mean absolute error of the seasonal naive baseline over the training period [20].

This formulation makes MASE intrinsically scale-independent. By dividing the forecast error by the historical baseline error of the same series, the original units of the data cancel out in the fraction [20]. Furthermore, because the denominator is a single quantity computed over the entire training set rather than a division performed at individual timestamps, zero observations pose no difficulty, although MASE remains undefined where the scaling denominator is itself zero, as occurs when the historical series is perfectly constant.

A MASE value strictly below 1.0 indicates that the forecasting model produces smaller errors than the naive baseline [20]. A value of exactly 1.0 indicates that the model performs as well as the naive baseline. Conversely, a value greater than 1.0 indicates that the forecasting model performs worse than simply repeating the historical seasonal value.

Chapter 3

Related Work

3.1 Sleeping Cell Detection Methods

The detection of sleeping cells is identified in the self-healing literature as a principal challenge [7,13]. Approaches differ in which part of the network produces the evidence the detection rests on, whether the user equipment, the cells surrounding the one under assessment, or the cell itself. A similar division appears in work on cell outage management in LTE, which lists the affected base station, its neighbouring base stations, the user equipment and the operation and maintenance system as sources of measurements for outage detection [1]. The following sections are organised by the source of that evidence, since it determines what remains available when a cell carries no traffic.

3.1.1 Detection from Device Measurements

The approaches in this section draw on what UE reports. Devices record the radio conditions they observe and the signalling events they generate, and these reach the network through Minimization of Drive Tests (MDT), a mechanism standardised by the 3rd Generation Partnership Project (3GPP) so that ordinary devices serve as measurement probes in place of manual field testing [9, 29]. Records are reported through whichever cell is serving the device and collected centrally for analysis.

Chernov et al. use the signalling events a UE produces while attached to the network, such as handover commands and radio link failures [10]. A long sequence of these events is cut into short segments, and each segment is described by how often each type of event appears in it. Detectors trained on error-free data flag the segments that depart from the normal pattern, and the failed cell is identified by comparing how often each cell appears among the flagged segments against how often it appears in normal operation. However, the authors mention that this method cannot detect failures in cells with very low user activity, as it depends on active signalling sequences to identify anomalies [10].

Chernogorov et al. also cut the event sequences into short segments, but describe each one by how often each pair of consecutive events occurs in it, rather than by how often each event occurs alone [9]. Counting pairs keeps the order in which events follow one another. A nearest-neighbour detector trained on error-free data flags the segments that sit furthest from normal behaviour. Each report is delivered over the device's active connection under immediate MDT, and names the target cell the device was moving towards. Counting those destinations across the flagged

segments shows which cell devices repeatedly failed to reach. A cell is labelled faulty if its count stands out against the network mean. The authors state that because the framework works from event sequences rather than numerical measurements, its performance does not depend on how users move or on changing traffic levels [9].

A second group works from the radio measurements that devices report rather than from the events they generate. Masood et al. use the reference signal received power (RSRP) and signal to interference plus noise ratio (SINR) a device observes from its serving cell and the three strongest neighbours [29]. The device reports these measurements through MDT, and the base station adds timing information before forwarding them to the management system. Each record also carries the position at which the measurement was taken. An autoencoder is trained on the measurements collected while the network is healthy, so it reconstructs familiar signal profiles closely and unfamiliar ones poorly. A measurement whose reconstruction error exceeds a threshold is flagged. The measurement values do not identify a cell, so the recorded positions are used afterwards to place the flagged measurements on the network topology. The cell that accumulates the most is identified as the one that failed. Asghar et al. train an autoencoder on healthy measurements in the same way, but use RSRP and reference signal received quality (RSRQ) and take reports periodically [3]. Each flagged measurement is mapped to its closest base station based on Euclidean distance. The base station accumulating the most is identified as the one that failed.

Zhang et al. work from RSRP and RSRQ for the serving cell and its three strongest neighbours, reported through immediate MDT to the management system [44]. They do not treat detection as a single fault state, but classify a cell as healthy, degraded, crippled, or catatonic. Because failures of each degree occur far less often than normal operation, a classifier trained on such data is biased towards predicting the healthy state. An adversarial autoencoder generates additional examples of the rarer degrees until the training set is balanced. A cost-sensitive support vector machine is trained on the balanced set, and assigns each cell to one of these classes from its own measurements.

Fortes et al. work from the received signal strength that devices report only for their serving cell [13]. The area around each cell in which its signal should be usable is calculated from the positions and transmit powers of the base stations. Each cell knows where its own devices are, so it can tell which of them are standing inside the area of another cell. If the target cell has failed, the devices it was serving attach to neighbours instead, and report poor signal because they are now served from further away. Each neighbour looks at what those devices report and computes an estimate of how likely a failure is. These estimates are then shared and combined to make a final decision. The authors state that narrowing the area makes a failure more visible but can leave too few reports to make a decision [13].

The approaches here require different things of the network. Those using signalling events need devices attached and generating activity, which Chernov et al. state directly [10]. Those using radio measurements do not need active sessions, since a device can measure a cell's signal whether or not it carries traffic, but they need a device present in the coverage area for any measurement to exist [13]. In every case the evidence originates at the device rather than at the cell, and all six approaches are evaluated on simulated networks.

3.1.2 Detection from Neighbouring Cells

Rather than drawing on what devices report, a second group of approaches observes the performance of the cells surrounding the one under assessment. When a cell stops serving, the devices in its coverage area attach elsewhere, and the evidence of the failure is produced by the neighbouring cells that take on that traffic.

Manzanilla-Salazar et al. address random access channel (RACH) failures, in which a cell continues serving connected users while rejecting new connections, so that its own counters remain populated and nothing appears wrong [28]. Devices unable to complete random access on that cell instead attempt the neighbour offering the next strongest signal. The additional attempts raise contention on that neighbour's random access channel, registering as increased collision probability and access delay, so the evidence of the failure appears at the surrounding cells rather than at the failed one. For each cell the authors identify the neighbours a device would turn to next if that cell failed, ranked by which it would choose first, and a classifier is trained on the counters of the cell together with those of its neighbours. The evaluation compares two traffic intensities in a simulated network, differing in the rate at which devices generate packets. The authors report that the level of traffic makes little difference to detection [28].

Zhang, Feng et al. address failures in dense deployments, where small cells are overlaid by macro cells covering the same area [43]. A handover is the transfer of a device's connection from one cell to another, and each cell records the number of handovers it receives. When a small cell fails, its connected devices are handed over to the overlapping macro cell without any drop in the signal they receive, so measurements taken by devices give no sign that anything is wrong. The failure shows instead as a sudden rise in handovers at the surrounding cells. A modified local outlier factor method flags the cells whose handover counts rise unusually. The flagged cells are neighbours of the failed cell, which is found as the one neighbour they all share. The authors report a false negative rate of three per cent, which they attribute to outages at cells carrying very low traffic that this approach cannot detect. They note that such outages have little effect on users or on overall network performance [43].

Detection in this family requires traffic to be displaced, and enough of it for the displacement to register at the neighbouring cells. Manzanilla-Salazar et al. report that the level of traffic makes little difference to detection, though the two levels compared differ in the rate at which devices generate packets rather than in the number of devices attempting access, while Zhang, Feng et al. report that outages at cells carrying very low traffic are not detected at all. Both evaluations use simulated networks, with the failure introduced by disabling the random access procedure or by attenuating transmit power.

3.1.3 Detection from Reported KPIs

This family draws on the counters a cell reports about its own operation, collected periodically at the network side. Because these are recorded automatically by standard logging systems, they offer continuous visibility into network conditions without requiring software at the device or measurement in the field. Detection in this setting

rests on establishing what a cell ordinarily reports and identifying departures from it.

Wu et al. present CellPAD, which treats anomaly detection as a regression problem. Six counters are monitored, covering user population, radio resource usage and transmission load, and predictors are fitted to estimate the value each should take, with daily and weekly structure captured through hour-of-day and day-of-week features. The prediction gives what the counter should read, and a fault shows as the observed value falling below it. This is expressed as the drop ratio, the difference between observed and predicted relative to the prediction. The drop ratios seen over the preceding week give a normal range. A drop is flagged when the ratio falls well below that range. The framework was evaluated on records from an operational LTE network. The dataset covers 12,463 cells over seventeen weeks, and anomalies were injected into a randomly selected sample of eighty cells by reducing counter values at selected hours by between thirty and one hundred per cent. The authors report that the counter recording user population fluctuates far more, relative to its typical level, between 23:00 and 06:00 than during the day [41].

Hawulte builds on this framework in a master's thesis, working from operational counter records taken from twenty sites of a Universal Mobile Telecommunications System (UMTS) network [16]. Two limitations of CellPAD are identified. The counters it monitors are driven by traffic, so during off-peak hours they sit so close to zero that a fault produces no departure large enough to register, and anomalies arising then go undetected. CellPAD also detects when two counters that normally move together stop doing so, but cannot say whether congestion or a fault is responsible. To approach the off-peak hours the thesis introduces the received total wideband power, a measurement of uplink noise and interference taken at the base station's own receiver that is present whether or not any device is transmitting. When a traffic counter falls while the number of active users is itself low, this measurement is used to separate the two cases: a normal level indicates an ordinary lull, and a degraded one indicates a fault [16].

Ming et al. detect sleeping cells in three stages [31]. Each counter is first separated with STL into seasonal, trend and residual components. The seasonal component is then removed, because the authors find that the regular daily pattern masks anomalies, and detection works on the trend and residual that remain. A normal range is set from the preceding days, and a cell is flagged if any of its values falls outside it. A second stage then discards cells whose values only spiked briefly, and a third removes those matching known causes other than a sleeping cell. The evaluation covers 21,657 cells and 38 counters at 15-minute granularity from an operational LTE network. The authors report that the seasonal component of counters such as random access attempts follows a stable daily cycle, but do not address what this implies for detection during quiet hours [31]. At an hour when a cell is normally quiet, a fault changes its values very little, because they are already close to zero. After the daily pattern is removed, the faulty values look almost the same as normal ones and stay well inside the normal range, so a fault arising at those hours is not flagged.

Le et al., in a preprint, combine a rule engine encoding domain expertise with components learned from unlabelled records and a classifier trained on injected anomalies [22]. One rule addresses the sleeping cell directly. It classifies a cell as sleeping

once its session setup success ratio, traffic and active user count have stayed near zero for longer than a set duration, a requirement intended to limit false alarms. Two hours is given as an example of this duration, which is presented as one of the heuristics operators configure in practice rather than derived from the data. The work draws on 194 cells over thirty days of operation from a 5G network, and because confirmed sleeping cell records were scarce, the supervised component was trained on anomalies injected into clean baseline data [22].

Sangaiah et al. take the counters as their primary automated source, but corroborate them with field drive tests and geographically mapped customer complaints [35]. Cells are first grouped by failure profile using unsupervised clustering, and a supervised classifier then predicts the cause. The work draws on one month of operational records covering fifty base stations of a Global System for Mobile Communications (GSM) network. To ensure the counters are statistically valid, the authors restrict their analysis to the busy hours, reasoning that the counters summarise network condition and hold only where the network is used enough [35]. Both corroborating sources depend on human activity, so neither is available during the hours the counters themselves are least informative.

These methods identify a fault by comparing what a cell reports against what it was expected to report. The comparison carries information whenever the expectation is substantial. When a cell is ordinarily quiet, the expected value is itself near zero, and a cell reporting nothing is then indistinguishable from one behaving as expected, which is the limitation Hawulte identifies and addresses by introducing a measurement that does not depend on traffic [16].

Table 3.1 summarises the works reviewed above by the evidence each uses and what that evidence depends on.

Reference	**Source**	**Evidence used**	**Depends on**
Chernov et al. [10]	Devices	Signalling events from the devices	Devices attached and active
Chernogorov et al. [9]	Devices	Ordered signalling events from the devices	Devices attached and active
Masood et al. [29]	Devices	Radio measurements from devices, with position	A device present in the coverage area
Asghar et al. [3]	Devices	Radio measurements from devices, with position	A device present in the coverage area
Zhang et al. [44]	Devices	Radio measurements from the devices	A device present in the coverage area
Fortes et al. [13]	Devices	Signal strength reported by devices now served by neighbours	Devices present in the failed cell's area
Manzanilla-Salazar et al. [28]	Neighbours	Random access contention at neighbouring cells	Access attempts displaced onto neighbours
Zhang, Feng et al. [43]	Neighbours	Handover counts at neighbouring cells	Traffic displaced onto neighbours
Wu et al. [41]	Own KPIs	Cell's own counters against a predicted value	An expected level clearly above zero

Reference	Source	Evidence used	Depends on
Hawulte [16]	Own KPIs	Cell's own counters and received total wideband power	An additional measurement at the cell's receiver
Ming et al. [31]	Own KPIs	Cell's own counters against a normal range from preceding days	An expected level clearly above zero
Le et al. [22]	Own KPIs	Cell's own counters held near zero	A persistence threshold, e.g. two hours
Sangaiah et al. [35]	Own KPIs	Cell's own counters, drive tests and user complaints	Busy-hour traffic

Table 3.1: Overview of reviewed sleeping cell detection work

3.2 Research Gap and Thesis Positioning

The approaches reviewed above differ in what they observe, and each depends on that observation being available. Those drawing on device reports require devices present in the coverage area, and Chernov et al. state that detection is not possible where the number of active users is low [10]. Those drawing on neighbouring cells require traffic to be displaced and enough of it to register, and Zhang, Feng et al. attribute their false negatives to cells carrying very low traffic [43], while Fortes et al. report that narrowing the area they assess can leave too few reports for a decision to be reached [13]. Those drawing on the cell's own counters do not require devices, since the counters are reported whatever the cell is doing.

The counters are therefore the only one of these three sources that remains available when a cell carries no traffic, and the reviewed works establish that their availability is not by itself sufficient. Detection in this family rests on forming an expectation of what a cell ordinarily reports and identifying a departure from it, and at the hours a cell is ordinarily quiet the expectation is itself close to zero. Hawulte reports that the observed and expected values are then too close for any departure to register, so anomalies arising in that period go undetected [16]. Sangaiah et al. reach the same position from the other direction, restricting their analysis to the busy hours on the grounds that the counters are reliable only when the network is used enough [35]. Le et al. address the case with a rule requiring the silent state to persist before a cell is classified as sleeping, so that transient dips do not raise alarms. The rule encodes what operators configure in practice, and the duration is fixed rather than derived from the behaviour of the individual cell [22].

The preceding sections leave two questions open within this family.

1. **Behaviour preceding a zero-traffic onset.** The reviewed approaches assess a cell during the period in which it reports nothing. None of them examine whether the interval leading up to that period differs between a cell that has failed and one that has simply run out of traffic to carry. This thesis examines that interval.

2. **Use of a per-cell expectation when the observation is zero.** Of the two works addressing this condition directly, one adds an additional measurement taken at the cell's receiver [16] and the other applies a rule with a wait of fixed duration [22]. Neither retains the expectation as the basis for judging the zero. This thesis retains the expectation and assesses the zero against the level of activity that cell would ordinarily reach at that moment.

This thesis works within the KPI counters supervision already collects, and does not claim to introduce a novel detection technique. Forecasting of performance counters is established in this literature and the work follows it. Its contribution lies in what the forecast is used to decide, and in examining an interval the reviewed work does not examine.

Chapter 4

Method

4.1 Research Design

4.1.1 Design Science Research Methodology

Design Science Research addresses problems whose resolution requires building artifacts. Hevner et al. describe it as building and assessing artifacts aimed at problems arising in organisations [17]. The artifact carries the contribution. Knowledge is produced by designing, and the artifact is judged against the problem it was built to address.

Advanced Cell Supervision cannot separate a healthy idle cell from a sleeping cell when no traffic is present, and the gap was identified through collaboration with the industrial partner before any solution was contemplated. Three features of the problem make design science research the appropriate framing. First, the outcome required is a method that did not exist in the supervision system. The contribution is therefore an artifact to be constructed. Second, the information that could support the decision was not known at the outset. The reports produced during a zero-traffic period do not distinguish the two conditions, and what could take their place had to be established before anything could be designed. Third, the data holds no record of confirmed faults. The method could not be assessed by comparison against known outcomes and an evaluation had to be constructed for it.

The artifact developed in this study is a detection method that assesses each zero-traffic observation against the activity a cell would ordinarily be carrying at that moment, together with the system through which the method is applied to reported counter data.

The research process follows the Design Science Research Methodology of Peffers et al., which organises the work into six activities [33]:
(1) Problem Identification and Motivation,
(2) Definition of Objectives for a Solution,
(3) Design and Development,
(4) Demonstration,
(5) Evaluation,
(6) Communication.

The methodology allows entry at different points in this sequence, and this study is problem-centered, having originated from a difficulty encountered in operation. Figure 4.1 shows the activities and the sections of this chapter that address each.

Under the knowledge contribution framework for design science research estab-

lished by Gregor and Hevner [14], the artifact developed in this study is classified as an *Improvement*.

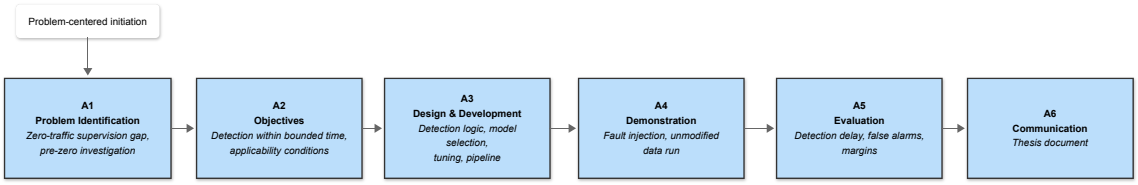


Figure 4.1: The six activities of the DSRM as applied in this study, adapted from Peffers et al. [33].

4.1.2 Mapping Research Questions to Study Activities

The objectives of the work are stated in Chapter 1, fulfilling Activity 2. The remaining activities are carried out within this chapter. Section 4.2 and Section 4.3 address Activity 1, establishing the problem and examining what the data contains. Section 4.4 addresses Activity 3, in which the detection method is designed and implemented. Section 4.5 addresses Activities 4 and 5, in which the method is applied to operational records and the quantities of interest are measured. Activity 6 is fulfilled by this thesis and the accompanying material supplied to the industrial partner.

Table 4.1 shows where each research question is addressed. Some sections contribute to more than one question. The outcomes are reported in Chapter 5 and interpreted in Chapter 6.

Table 4.1: Research questions mapped to the activities addressing them

Research Question	Activities	Sections
RQ1: What AI-based approach can distinguish sleeping cells from healthy idle cells during zero-traffic periods in cellular networks?	Problem Identification, Design and Development, Demonstration, Evaluation	4.2, 4.3, 4.4, 4.5
RQ2: What features or patterns extracted from cellular network data can effectively differentiate healthy zero-traffic cases from fault-related zero-activity conditions?	Problem Identification, Design and Development	4.2, 4.3, 4.4.1
RQ3: What detection performance does the proposed approach achieve in distinguishing healthy idle cells from sleeping cells during zero-traffic periods?	Demonstration, Evaluation	4.5

4.1.3 Alternative Methodologies

Experimental research was considered as the primary methodological alternative for this study. The research questions ask what separates the two conditions and what

detection performance is achieved, and questions of that form are ordinarily addressed by comparing treatments under controlled conditions [40].

Experimental research was rejected for three reasons.

- **Artifact Construction.** The outcome of the work is a detection method that had to be constructed, whereas an experiment compares treatments that already exist in order to establish an effect.
- **Absence of Ground-Truth Labels.** The data holds no labelled occurrences of sleeping cells. Detection correctness is therefore not observable as a dependent variable, and a controlled comparison of detection performance cannot be carried out.
- **Absence of a Deployed Alternative.** No detection for this condition exists in the supervision system the work extends, so no deployed alternative was available to compare against.

Case study research, which studies a phenomenon as it occurs in a real setting [34], would not suit this work either, since no sleeping cell was confirmed in the data and the work had to build a method. Design science research was therefore adopted as the methodology for this work. As with any design science study, the outcomes describe the artifact as built and evaluated in one setting, and how far they generalise is discussed in Section 6.5.

4.1.4 Literature Survey Protocol

The literature reviewed in Chapter 3 was identified through a structured, non-systematic search driven by the research questions. The search was conducted in Google Scholar, IEEE Xplore and the BTH library search service (Summon), combining domain terms such as “sleeping cell”, “zero traffic”, “self-healing”, “self-organising networks”, “cellular networks” and “LTE” with detection terms such as “detection”, “anomaly detection” and “outage detection”. No restriction on publication year was applied, and a work was retained where it helped answer the research questions. Backward referencing was used to broaden the coverage. The retained works were then grouped by the source of evidence each draws on. The thesis aims to design a detection method rather than to survey the field, so a full systematic literature review, as described by Kitchenham and Charters [21], was not conducted.

4.2 Problem Identification (Activity 1)

4.2.1 Advanced Cell Supervision

Automated supervision infers cell health from the KPI counters described in Section 2.2.

Advanced Cell Supervision (ACS) is the supervision capability deployed by the industrial partner. It operates on KPI counters and reliably identifies faulted cells across a broad range of fault conditions when the cells are engaged in serving users.

The internal detection mechanisms, configuration, and deployment arrangement of ACS are proprietary and lie outside the scope of this work. However, the specific property relevant to this thesis is that its reliance on active traffic creates a known blind spot for sleeping cell detection during zero-traffic scenarios.

Every inference ACS draws about cell health requires the counters to carry positive values. A cell that has failed while idle remains unreported and raises no alarm.

4.2.2 Supervision Data

The data used in this study consists of KPI counters reported by cells in an operational LTE network. The corpus comprises 1024 distinct cells drawn from three regions, distributed as shown in Table 4.2. No cell appears under more than one region. Site identifiers, cell identifiers, and the geographic location of the network are anonymised throughout this thesis, and the calendar period covered by the observations is not disclosed.

Table 4.2: Distribution of cells across regions.

Region	**Cells**
A	234
B	306
C	484
Total	1024

Nine counters are available for each cell. The counters follow the sequence by which a device obtains service from a cell. A device seeking a connection first performs random access on the random access channel (RACH STP Attempts, RACH STP Fails), then proceeds to radio resource control connection setup (RRC Conn Setup Attempts, RRC Conn Setup Successes), and once connected generates the call attempts and downlink traffic that the cell serves. Attempts, failures, and successes at these stages are recorded as absolute counts. The remaining counters are proportions derived from such counts, expressing success rates and the relationship between consecutive stages of the access procedure. Table 4.3 lists the counters with their units.

Counter values are timestamped and reported at a fixed interval of 15 minutes, so each cell contributes 96 observations per day for every counter. The interval is uniform across all cells and throughout the observation period. The period over which observations are available differs between regions and between cells within a region, and the spans relevant to each part of this study are stated where those data are used.

The corpus carries no annotations of any kind. It contains no labels identifying faults or outages.

Table 4.3: KPI counters available for each cell.

Counter	Description	Unit
RACH STP Attempts	Random access channel setup attempts received by the cell	Count
RACH STP Fails	Random access channel setup attempts that failed	Count
RRC Conn Setup Attempts Total	Radio resource control connection setup attempts initiated at the cell	Count
RRC Conn Setup Successes	Radio resource control connection setup attempts completed successfully	Count
Call Attempts	Call setup attempts processed by the cell	Count
DL Traffic Volume	Downlink data volume served by the cell	MB
RACH CBRA Success Rate	Proportion of contention-based random access procedures that succeeded	%
RRC-to-RACH Attempt Ratio	Connection setup attempts relative to random access attempts	%
Cell Availability	Proportion of the reporting interval during which the cell was operationally available	%

4.2.3 Zero-Traffic Conditions

Across the observation period, cells regularly report intervals in which no activity occurs. Within such an interval, every counter that records the volume or outcome of a procedure reads zero, since no random access attempt, connection setup, or call setup was registered and no downlink data was served.

The derived proportions likewise report zero. Cell availability continues to report the cell as operationally available throughout. A cell in this state therefore reports that it is available but has handled nothing. This condition is referred to in this thesis as zero traffic, and an interval of consecutive reports in this state is referred to as a zero-traffic period.

A cell may have no users to serve, which is an ordinary and expected condition when no one in its coverage area is trying to use it. Alternatively, a cell may have failed in the manner described in Section 2.3, remaining operational from the perspective of the monitoring system while serving no users and raising no alarm. These two circumstances demand entirely different responses from an operator. But the counters record only what a cell handled, and a cell that handled nothing writes the same record either way. No quantity computed from the reports within a zero-traffic period can therefore separate the two.

4.2.4 Absence of Fault Labels

The corpus holds no record of which cells failed at any point in the observation period, as noted in Section 4.2.2. Studies working with operational network data have reported the same situation and developed detection methods that do not depend on confirmed fault cases [41]. This work proceeds on the same basis, using the observations alone.

No zero-traffic period in the corpus can therefore be checked against a recorded outcome. Detection performance cannot be counted directly from the data, and the evaluation is constructed accordingly, as described in Section 4.5.

4.3 Exploratory Analysis of Pre-Zero Behaviour (Activity 1)

Section 4.2.3 established that the reports produced within a zero-traffic period carry no information about what caused it. This does not preclude such information existing elsewhere, and the interval immediately preceding the onset is the most direct place to look for it. A cell that stops serving users because it has failed may pass through that transition differently from one that simply runs out of traffic to carry, and any such difference might be visible in the reports leading up to the onset. This section describes the analysis carried out to find out whether it does.

The analysis uses eight of the nine counters described in Section 4.2.2. Cell availability is excluded, as it reports the cell as fully available on more than 99% of all records and consequently offers no basis on which records could be distinguished from one another.

4.3.1 Window Extraction

A zero-traffic onset is the first timestamp at which a cell enters the state described in Section 4.2.3, where all eight counters simultaneously report zero. This marks the point from which the cell reports no activity of any kind.

The interval preceding each onset is examined as a window of four consecutive reports, covering one hour at the fifteen-minute reporting interval. Four is the shortest window that permits the intended characterisation of the interval. Both the trend of a signal and its variability require at least three points to be estimated meaningfully, and a fourth allows the window to be divided into two halves for comparison.

Three conditions determine whether a window is retained. The cell must report at least four records showing activity before the window begins, so that the window is preceded by demonstrable operation rather than beginning at the start of the cell's series. No record within the window may report zero across all counters, so that the window describes behaviour leading toward the onset rather than an interval already at rest. The first and last records of the window must fall within sixty minutes of one another, which admits windows containing a single missing report while excluding those spanning a longer gap.

Applying these conditions across the corpus yields 37,585 windows. These come from 741 cells across the three regions, each contributing at least one window that

met all three conditions.

4.3.2 Normalisation

Cells differ substantially in the volume of traffic they carry, so the same counter value can represent ordinary operation at one cell and unusual activity at another. Before any comparison across cells is possible, each cell's records must be expressed relative to that cell's own behaviour.

That behaviour has to be characterised from the cell's active periods rather than from its full history, which would otherwise include records without downlink activity and pull the reference statistics toward zero. A cell reports positive downlink volume only when it is operational and actively carrying traffic, which makes this counter a direct indicator of the periods sought. Every record in which a cell reported positive downlink volume is therefore taken as its reference set.

The six counters that record counts and volume are first transformed as $\log(1+x)$, which compresses the wide range these counters span while remaining defined at zero. The two ratio counters are left untransformed, as they are already bounded and a log transformation is not appropriate for them.

Each counter is then standardised against the median and median absolute deviation of that counter across the reference set. These are used in place of the mean and standard deviation because operational counter data contains occasional extreme values, and the mean and standard deviation are both sensitive to them [23]. If a cell's median absolute deviation is zero, a value of one is substituted so that the calculation can proceed.

4.3.3 Feature Construction

If a fault-induced transition differs from a natural one, the form that difference takes is not known in advance. Traffic might slide away steadily, become erratic, fall sharply from a peak, or shift partway through the interval, and any of these could distinguish the two transitions. The window is therefore described through four measures, each capturing a different aspect of how the counter behaved across it. These were chosen to cover the plausible forms a difference might take. The four measures applied to each counter are listed in Table 4.4.

Table 4.4: Measures computed for each counter over the window.

Measure	**What it captures**
Slope	The direction and rate of change across the window
Volatility	How erratic the counter is within the window
Peak-to-final drop	How far the counter fell from its highest value to its last
Half-to-half difference	The shift in behaviour between the first and second halves

Applied across the eight counters, these give 32 features. Five further features describe relationships between counters rather than the behaviour of any one.

Three of these are taken from the last report in the window, immediately before the onset. The first is the proportion of random access attempts that failed, and the second the proportion of connection setup attempts that succeeded. Unlike the other features, both are computed from raw counts rather than normalised values, since a proportion is already comparable across cells of different size. The number of attempts is floored at one so that both remain defined when no attempts were recorded. The third is the contention-based random access success rate at the final report, whose change across the window is already captured by the four measures. The remaining two compare the trends of related counters, contrasting random access attempts with the contention-based success rate, and connection setup attempts with connection setup successes. Each is the difference between their normalised slopes.

This gives 37 features in total, and none was discarded. The analysis has no prior expectation of which description might carry a difference, so discarding any of them would risk removing the one that does. The window is short enough that all can be retained without difficulty. The complete list of features is given in Appendix A.1. Before detection, each feature is standardised across all 37,585 windows so that the detectors treat every feature on the same scale.

4.3.4 Anomaly Detection

The question posed at the start of this section is whether the interval preceding a zero-traffic onset differs when the onset is caused by a fault. This interval is now represented as 37,585 windows, each described by 37 features. If fault-induced transitions differ from natural ones, that difference would appear as a subset of windows standing apart from the remainder within the feature space.

No record identifies which onsets were caused by faults, so neither group can be specified in advance and no model can be trained to separate them. Instead, the analysis looks for structure a detector can find without supervision.

Each detector partitions the windows by construction, selecting a fixed proportion as candidates whether or not any meaningful division exists. The presence of candidates is therefore uninformative in itself. The analysis examines whether the selected windows share a consistent basis for having been selected, and whether that basis is pronounced enough to classify an individual window. A division without a consistent basis would not support detection, as no property would remain by which a future window could be recognised.

A window can be unusual in two distinct senses. In the first, it lies far from the data as a whole. In the second, it occupies a sparse region relative to its neighbours, even if it sits near the centre of the distribution. Neither sense has priority here, so both are examined. Isolation Forest identifies windows in the first sense [24] and Local Outlier Factor identifies windows in the second [8]. Both are applied independently to the same feature matrix. If two methods resting on different principles agree on which features drive their selections, that agreement indicates the pattern is present in the data rather than produced by either mechanism.

Isolation Forest is applied with 200 trees, above the library default of 100 adopted by Liu et al., who report that average path lengths converge well before that num-

ber [24]. Because the partitioning is random, running the detector again would rank individual windows slightly differently, so the comparison in Section 5.1 uses averages over several thousand windows. A fixed random seed makes the results reproducible.

Local Outlier Factor is applied with 20 neighbours, the default of the implementation used. Breunig et al. recommend a lower bound of at least 10 to remove statistical fluctuation from the local density estimates, and report that values between 10 and 20 work well across the datasets they examine [8]. Their upper bound is set by the largest group of points a user is willing to treat as local outliers, which cannot be fixed in advance here, since the size of any fault-induced group is unknown.

Each detector ranks every window by its own measure of standing apart, and a threshold determines how many of the highest-ranked windows are taken as candidates. The highest-ranked 10% of windows are taken as candidates from each detector.

4.3.5 Comparison Procedure

Each detector selects a fixed proportion of windows as candidates, so the remaining question is whether those candidates differ from the rest in any consistent way. This is examined feature by feature. For each of the 37 features, the mean value across the candidates is compared with the mean across the remaining windows, and the difference between the two is recorded. The feature matrix was standardised before detection, so each difference is expressed in standard deviations and the 37 are directly comparable. A larger difference indicates a feature on which the two groups are further apart.

The comparison is computed separately for each detector, using that detector's own candidates, so that two independent rankings are produced. The 10% threshold gives enough candidates for the group means to be stable, while keeping the comparison to the windows each detector ranked highest.

The Mann–Whitney U test is applied to the ten highest-ranked features from each detector, to establish whether the differences could have arisen by chance. Section 5.1 reports what the comparison found.

4.4 Detection Method Design and Implementation (Activity 3)

Section 4.2.3 established that the reports produced during a zero-traffic period do not distinguish a failed cell from an idle one, and the analysis described in Section 4.3 found no such distinction in the interval leading up to the onset (Section 5.1). The counters, however, carry more than their values. A cell observed over a period of weeks also builds a record of when it ordinarily carries traffic, which remains available even when the cell reports nothing. A zero observed at a time when the cell ordinarily carries traffic is not the same observation as a zero at a time when it ordinarily does not, although the two are identical in the counters themselves. This section describes a detection method built on that distinction, and the system through which it is implemented.

4.4.1 Regularity in Cell Traffic

Every counter record is timestamped, and each cell has several weeks of records. Every zero therefore occurs at a known time of day, for which the cell's own history shows its ordinary behaviour.

Counter values recorded at fixed intervals over an extended period form a time series, and time-series methods look for structure that appears across time. Time-series methods have been applied to network counter data in previous work [41], including in the detection of sleeping cells [16]. Whether the series in this corpus carry such structure is examined below.

The industrial partner's domain experts identified fifty cells in the corpus described in Section 4.2.2 as having operated normally throughout the observation period, and these are used to characterise the structure of the counter series. This identification rests on their operational knowledge of the network rather than on any information in the corpus. Each cell contributes 45 days of records at the fifteen-minute reporting interval. The characterisation is carried out on the RRC connection setup attempts counter, which records the volume of connection requests a cell handles. The properties reported are those of this set of cells over this observation period.

The recorded series show a repeating pattern across the observation period. Activity rises through the day, reaches a peak, and falls to zero or near-zero overnight, and this shape recurs from one day to the next. Diurnal variation of this kind is reported in cellular traffic more broadly [42] and in cell-level performance counters including RRC requests [16], though its shape differs between areas of different use.

Figures 4.2 and 4.3 show the series for two cells operating at different traffic levels, over the full observation period and over a single week.

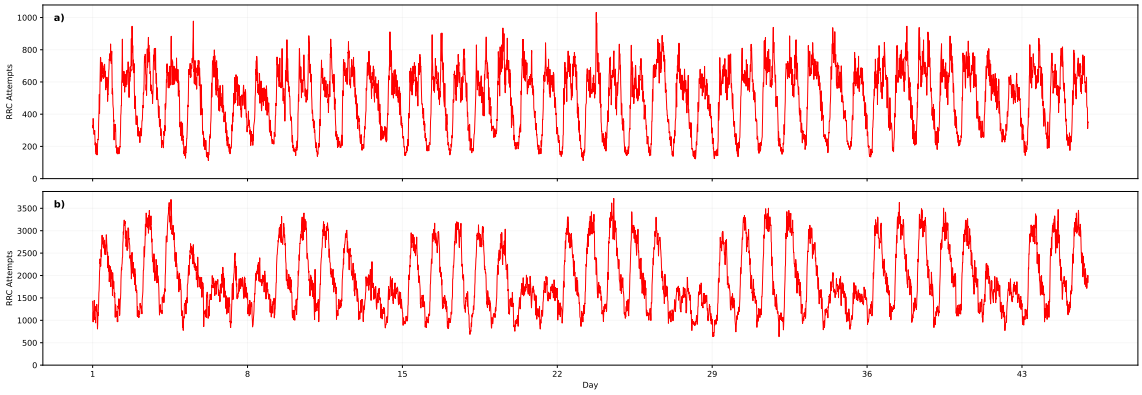


Figure 4.2: Recorded RRC connection setup attempts over the full observation period of 45 days, for (a) a cell with median non-zero value 536 and (b) a cell with 1754.

Decomposition separates an observed series into the components it contains, allowing each to be examined on its own. Multiple seasonal-trend decomposition using loess was applied with daily and weekly periods, corresponding to 96 and 672 reporting intervals. Figures 4.4 and 4.5 show the resulting components for the same two cells. The series resolve into a daily component with a consistent shape across days, a weekly component varying in amplitude between days of the week, a slow-moving trend, and a remainder that is smaller than the seasonal parts.

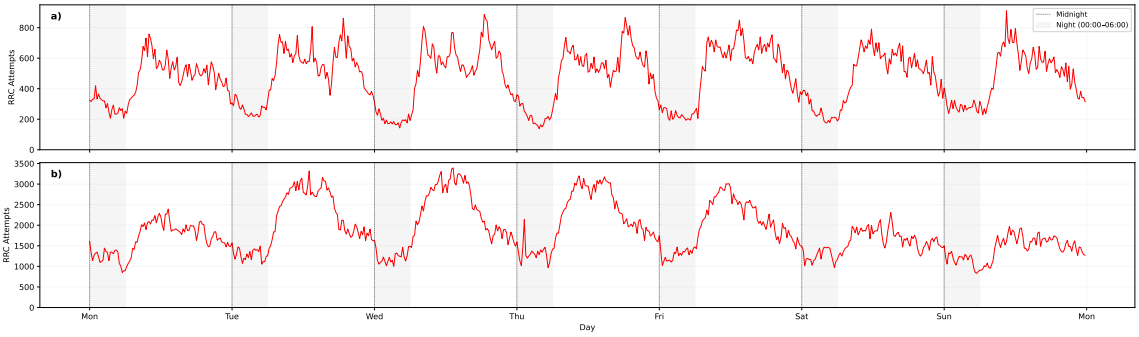


Figure 4.3: The same two cells over a single week.

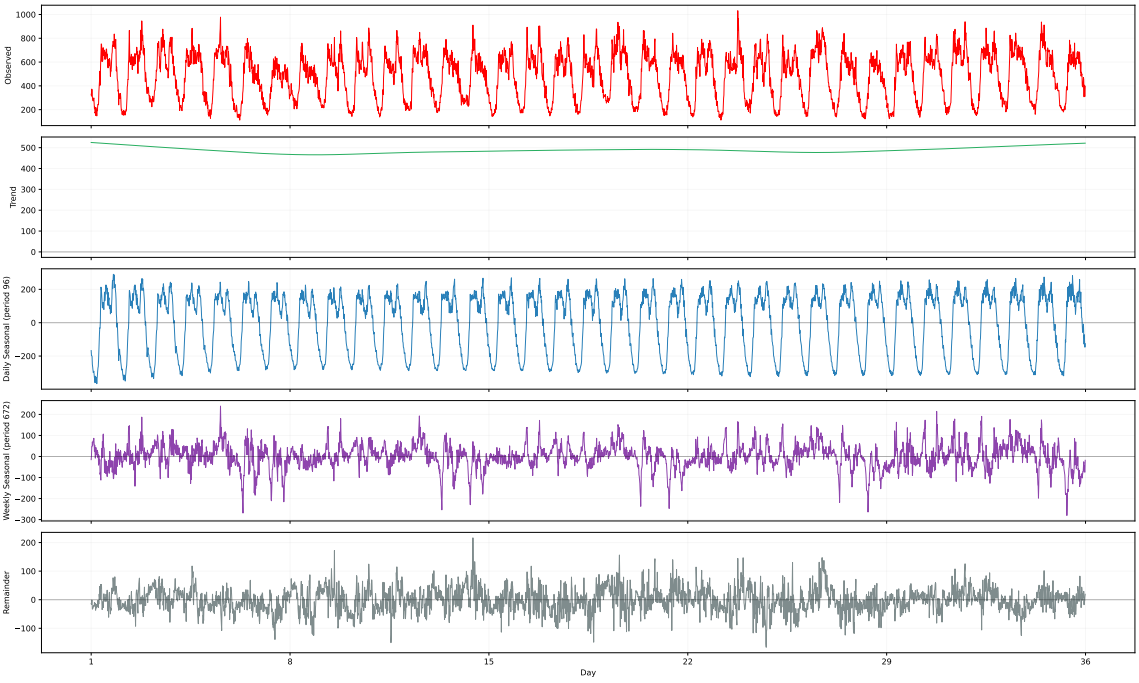


Figure 4.4: Multiple seasonal-trend decomposition for the first cell, showing the observed series, trend, daily component, weekly component, and remainder.

The strength of each component was quantified across the fifty cells using the seasonal strength measure, which gives the proportion of detrended variation attributable to the seasonal component. Table 4.5 reports the mean strength for each period and the number of cells exceeding a strength of 0.6, a value used here as a descriptive cut rather than a decision boundary. Daily structure exceeds this level in the large majority of cells, and weekly structure in a smaller number. The decomposition covers roughly the first five weeks of records.

The cells differ substantially in the volume of traffic they carry. Across the fifty cells, the median non-zero value of the counter ranges from 25 to 4707, so the level that constitutes ordinary activity for one cell may represent an extreme for another.

Two observations follow. The series carry structure that recurs on a daily cycle and, in a smaller number of cells, on a weekly one, so the level of activity a cell reaches at a given time can be described from its own history. That level differs

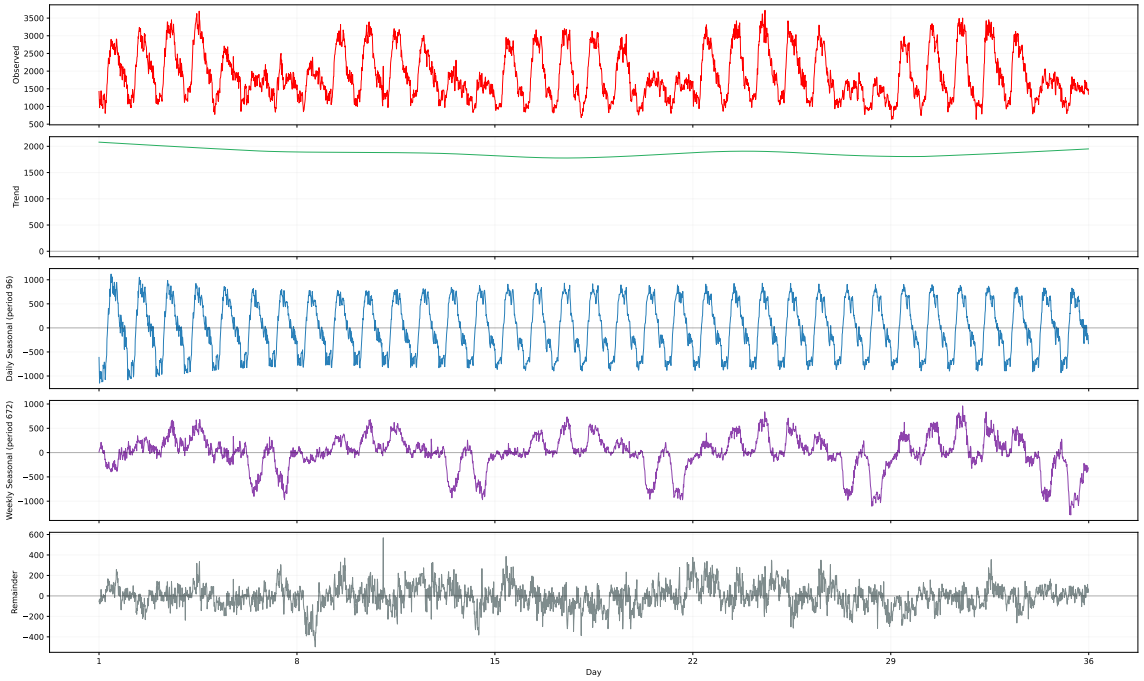


Figure 4.5: Multiple seasonal-trend decomposition for the second cell.

Table 4.5: Seasonal strength across the fifty cells

Period	Mean strength	Cells above 0.6
Daily (96 intervals)	0.760	45 / 50
Weekly (672 intervals)	0.576	18 / 50

between cells across the range observed here, and a description drawn from one cell does not transfer to another. Together, these make it possible to ask, for a cell at a particular moment, whether it would ordinarily be carrying traffic.

4.4.2 Detection Principle

Deciding whether a cell would ordinarily be carrying traffic at a given moment requires two things to be established for that cell. The first is a description of the level of activity the cell reaches at any given time, derived from its own history. The second is the level at which that cell counts as active at all. Both are expressed in terms of a measured quantity, so the first step is to determine which of the reported counters serves as that quantity.

Not all nine counters described in Section 4.2.2 are suited to this purpose. Cell availability reports the cell as fully available on more than 99% of records and so provides no basis for distinguishing one moment from another. The two ratio counters are proportions rather than measures of volume. Under zero traffic they read the same whether the cell is healthy or has failed. The count of failed random access attempts records errors rather than activity, and a high failure count does not imply that the cell should be active.

Five counters remain, recording call attempts, downlink traffic volume, random access attempts, connection setup attempts, and connection setup successes. All five arise from users being served by the cell, and they consequently rise and fall together and reach zero together when the cell carries no traffic.

Because the five move together, one of them describes the cell's activity rhythm as well as all five would, and modelling each separately would enlarge the design without adding information. The RRC connection setup attempts counter is used as the single activity indicator. It records connection establishment requests handled by the cell, and has been used as an activity indicator in prior work on network monitoring [16]. The remaining four counters are not used by the method.

The way the expected level of the indicator is obtained for a given moment is described in Section 4.4.3. The second quantity is the level at which the cell counts as active. This is taken as the median of the cell's non-zero values of the indicator over its training period. Only non-zero records contribute, so the level reflects what the cell does when it is serving users rather than how often it is idle. The median is used rather than the mean because traffic is right-skewed, and busy-hour peaks would draw a mean above the cell's typical active level. Section 4.4.1 showed that this level differs between cells by a wide margin, so it is computed separately for each cell from that cell's own history and held fixed thereafter.

Because the method operates on this single indicator, a zero-traffic onset is identified here by the indicator falling from a positive value to zero, rather than by the simultaneous condition on all counters used in Section 4.3.1. That condition is stricter, as it requires every counter to be zero, so its onsets are a subset of those found from the indicator alone. The counters reach zero together, so the two are expected to coincide in most cases.

4.4.3 Detection Logic

The expectation against which a zero is assessed comes from a model fitted to each cell's own history of the activity indicator. The model is fitted before the system is put into use. From any point in time, it can then forecast the indicator one day ahead, a horizon of 96 reporting intervals. One day is used because the structure described in Section 4.4.1 repeats on a daily cycle, so a cell's next active period falls within it, while a longer horizon would forecast further ahead at greater error [19]. Section 4.4.4 describes which model is used and how it was chosen.

A forecast is generated at the moment the indicator falls from a positive value to zero. Any negative values it contains are set to zero, since the model can produce them algebraically while the counter itself cannot fall below zero. The first position in this forecast is the onset timestamp, so the value there is the activity the cell would normally have had at that moment.

The forecast value at the onset is compared against the cell's activity threshold, the median of its non-zero values, and the two cases that result differ in what the observation can settle on its own. A cell that would ordinarily have been active contradicts its own established behaviour by reporting zero, and nothing further is required. A cell that would ordinarily have been quiet reports zero whether it has no users to serve or has failed, so the observation settles nothing and the assessment cannot be completed at the onset.

Case A. The forecast at the onset exceeds the threshold.
The cell would ordinarily have been carrying traffic at that moment. An alarm is raised at the onset itself, with no detection delay, and the episode is closed.

Case B. The forecast at the onset does not exceed the threshold.
The episode is held open, and a timer begins at the onset and advances through each subsequent reporting interval in which the indicator remains at zero.

An open episode needs a point at which the waiting ends. That point must be a moment when the cell is expected to be active, since silence then can no longer be put down to idleness. Two rules determine where it falls.

First crossing.
The deadline is the first timestamp within the forecast horizon at which the forecast exceeds the threshold, this being the earliest point at which the cell should next be active.

Peak backstop.
A forecast that never exceeds the threshold within the horizon gives no crossing, so the deadline falls instead at the timestamp carrying the highest forecast value in the remainder of the horizon, excluding the onset. This marks the point at which the expectation of activity is strongest and continued silence is least consistent with ordinary operation.

The episode then closes in one of two ways. If the indicator returns to a positive value before the deadline is reached, the cell has resumed serving traffic. The episode is closed as an ordinary idle period and the timer is reset. If the deadline is reached with the indicator still at zero, the cell has stayed silent at a time its own history says it should have been active. An alarm is raised, and the detection delay is the interval between the onset and the deadline. If the indicator later falls to zero again, this is treated as a new onset and assessed in the same way. Figure 4.6 illustrates Cases A and B across the daily cycle, and Table 4.6 summarises the outcomes.

Table 4.6: Detection outcomes by condition at the onset

Condition at the onset	Outcome
Forecast above threshold	Alarm at the onset
Forecast at or below threshold, traffic resumes before deadline	No alarm
Forecast at or below threshold, zero persists to deadline	Alarm at the deadline

Three properties follow from this construction. First, the assessment is specific to each cell. Its threshold, forecast and deadline all come from that cell's own history rather than from a standard shared across the network. Second, for cells assessed in this way, the delay before an alarm is bounded by one daily cycle, or 23 hours and 45 minutes, since the peak backstop ensures a deadline is always found within

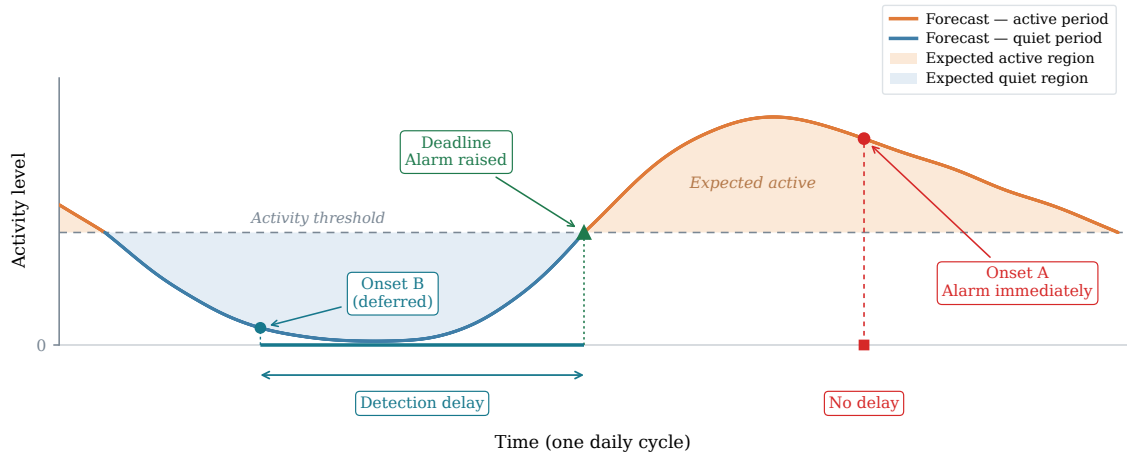


Figure 4.6: Assessment of two zero-traffic onsets across a daily cycle, shown schematically. The horizontal segments at zero show the observed indicator for each onset. Onset A falls where the forecast exceeds the threshold and raises an alarm at once. Onset B falls below it, and the deadline is set by the first crossing.

the forecast horizon, even where the forecast never crosses the threshold. Third, the assessment is deterministic, so assessing the same onset again always gives the same outcome.

The assessment works only for cells whose history yields a usable forecast. Section 4.4.5 sets out the conditions a forecast must meet, and how cells that do not meet them are handled.

4.4.4 Forecast Model Selection

Section 4.4.3 described the assessment without specifying what produces the forecast. Four properties of the data determine which models are suitable. Section 4.4.1 established a daily cycle of 96 reporting intervals and a weekly cycle of 672, and both must be captured simultaneously. That section also showed that activity levels differ between cells and do not transfer, so one model is fitted per cell from that cell's own history.

That history spans 45 days, giving 4,320 observations. The indicator is a non-negative count reaching zero during idle periods, which makes percentage-based error measures unavailable. Three families of model are ruled out by these properties.

Seasonal ARIMA.
Excluded because it accommodates only one seasonal period, so it cannot capture the daily and weekly cycles together [19].

Standalone exponential smoothing (ETS).
Excluded because its seasonal component keeps a separate value for every step of the cycle, which for the 96-interval daily cycle is too many to estimate reliably [12, 19]. Exponential smoothing still returns below as the second stage of one candidate, where it works on a series from which the seasonal components have already been removed.

Neural forecasting approaches.
Excluded because each cell's history is too short. A model is fitted independently to each cell, and a few thousand observations are not enough to estimate the many parameters that neural approaches involve. Comparative evaluation across large collections of individual seasonal series has likewise not found an advantage for these approaches over statistical methods at this scale [26].

The three candidates are the representatives of the three approaches to multiple seasonality described in Section 2.5.3, so the comparison spans distinct mechanisms.

MSTL+ETS, representing loess decomposition.
MSTL separates the series into daily and weekly seasonal components, a trend, and a remainder, with no ceiling on the seasonal period and with the seasonal shapes permitted to change gradually over the series [6,11]. Exponential smoothing forecasts the deseasonalised part, and the seasonal components are extended by repeating their last full cycle.

TBATS, representing trigonometric state-space modelling.
Both seasonal periods are modelled simultaneously as time-varying trigonometric terms within a single state-space framework, with all components estimated jointly [12]. The Box–Cox transformation was disabled, as it requires strictly positive values and the indicator reaches zero.

Prophet, representing Fourier regression.
The series is represented as a piecewise linear trend, whose slope changes at automatically placed changepoints, together with each seasonal period as an independent Fourier series, estimated within a regression framework [38]. Yearly seasonality was disabled, the observation period being far shorter than a year.

Apart from these two adjustments, disabling the Box–Cox transformation in TBATS and yearly seasonality in Prophet, each candidate was run at its default configuration. Forecast accuracy is measured by the mean absolute scaled error, or MASE, in which a model's mean absolute error on held-out data is divided by the mean absolute error of a seasonal naive baseline computed for the same cell [20]. Because the divisor is specific to the cell, MASE is free of scale and can be averaged across cells whose traffic volumes differ widely. A MASE below one indicates that the model improves on the baseline for that cell, and a value at or above one indicates that it does not.

The naive baseline repeats the value observed one full cycle earlier. There are two options for that cycle, the daily period of 96 intervals and the weekly period of 672. The choice between them was made before any model was fitted. Both were evaluated over a common window of the training data so that each had the same length of lookback available, and the daily period produced the lower average error across the fifty cells. It was therefore adopted for all cells and all models. Each cell's naive error was computed once over its full training range and used as that cell's fixed divisor.

The comparison used the fifty cells described in Section 4.4.1, under the protocol given in Table 4.7. The larger portion of each series is given to training because estimating a weekly cycle of 672 intervals requires a substantial run of history. Forecasts

were compared as produced, without the clipping applied in the detection system, so that forecasting capability was measured directly and identically for all three candidates.

Table 4.7: Comparison protocol

Element	Setting
Cells	50
Training	First 35 days, 3,360 intervals
Held out	Following 10 days, 960 intervals
Fitting	Once per cell, no refitting
Forecasting	Single forward call over the held-out period
Configuration	Defaults, apart from the two adjustments above
Metric	MASE against the lag-96 naive baseline
Criterion	Lowest mean MASE across the fifty cells, fixed before the comparison

Prophet records the lowest mean MASE of the three candidates, satisfying the criterion fixed before the comparison. It also reaches more cells than either alternative and is the only candidate whose median MASE falls below one, meaning that on the typical cell it improves on the naive baseline. Table 4.8 reports the outcome.

Table 4.8: MASE across the fifty cells

Candidate	Cells with MASE < 1	Mean MASE	Median MASE	Highest MASE
MSTL+ETS	21 / 50	1.463	1.048	16.854
TBATS	22 / 50	1.336	1.072	4.586
Prophet	29 / 50	1.114	0.945	2.745

The averages in Table 4.8 make the gap between the models look larger than it is on individual cells. When the best model is counted for each cell, Prophet is best on 20 cells, TBATS on 19 and MSTL+ETS on 11. Prophet and TBATS therefore win almost equally often. The difference in their averages comes largely from the cells where TBATS performs poorly. Its highest MASE is 4.586, compared with 2.745 for Prophet. MSTL+ETS performs worst, with a highest MASE of 16.854, more than ten times the error of the naive baseline.

The candidates therefore differ mainly in how many cells they reach and in how badly they fail on the rest. These matter for detection, since the forecast sets the deadline at which an alarm is raised, and a bad forecast places that deadline in the wrong position.

Taking the best model for each cell individually, 41 of the 50 cells improve on the naive baseline under at least one candidate. Prophet reaches 29 of these, so selecting it as a single model class leaves 12 that are not reached by Prophet but are reached by another candidate. Selecting a model for each cell was not adopted, for two reasons. First, the choice would be made on the held-out days and then assessed on those same days, with no unseen data left to check it. Second, the best model

for a cell could not be identified in advance. The models did well on different cells, with no apparent reason why a given model suited a given cell. The per-candidate breakdown behind these figures is given in Appendix A.3.

The remaining nine cells are reached by none of the three models. The system therefore does not assume that every cell can be assessed by forecast, and Section 4.4.5 sets out how cells are separated on this basis.

One further comparison confirms that both seasonal periods contribute. Running MSTL+ETS with the daily period alone reaches 16 cells, and with both periods 21. That candidate was used for this comparison because its decomposition is defined per period, so one period can be removed directly.

4.4.5 Model Configuration and Applicability

The comparison in Section 4.4.4 ran every candidate at its default configuration, since its purpose was to select between mechanisms rather than between tuned models. With the mechanism settled, the selected model can be configured for this data.

Prophet exposes several parameters that control how closely the fitted model follows the training series. Three of them were varied here, the two prior scales governing regularisation and the seasonality mode. Table 4.9 gives each parameter, what it controls, and the values examined.

Table 4.9: Parameters varied and values examined

Parameter	Controls	Values examined
Changepoint prior scale	How freely the trend may bend	0.01, 0.05, 0.1, 0.3, 0.5
Seasonality prior scale	How large the seasonal swings may become	1.0, 10.0, 20.0
Seasonality mode	Whether the seasonal effect is constant or scales with level	Additive, multiplicative

The values examined for the changepoint prior scale run from 0.01 to 0.5, the upper end of the interval given in the model's documentation and covering a trend from near-rigid to highly flexible, with the default of 0.05 included [30]. Values below 0.01 were not examined. Since 0.01 already produces a nearly linear trend, further reduction would be expected to change the fitted trend very little. For the seasonality prior scale, the default is 10.0. The value below it, 1.0, tests whether tighter regularisation helps avoid overfitting on a short training series. The value above it, 20.0, tests whether allowing larger seasonal swings helps. Values below 1.0 were not examined. Section 4.4.1 established a strong daily seasonal component across these cells, and stronger regularisation than 1.0 would be expected to suppress it. The two seasonality modes are the only ones the model provides. The Fourier orders that determine the sharpness of each seasonal shape were left at their defaults, the search being limited to the two prior scales and the seasonality mode.

The three parameters can interact, so every combination was fitted and evaluated, five changepoint values by three seasonality prior values by two seasonality modes, giving thirty configurations in all.

Choosing among them requires a period of data on which configurations can be compared, and that period cannot be the one reserved for evaluating the outcome. The training portion of each series was therefore divided further. The first 28 days were used to fit each configuration, and the following 7 days to compare them. Once the configuration with the lowest mean MASE on the seven validation days had been identified, the model was refitted on the full 35 days and evaluated once on the 10 days held out throughout, which are the same days used in the comparison of Section 4.4.4. Parameter selection therefore made no use of the period on which the outcome is reported.

The configuration selected sets the changepoint prior scale to 0.01, the seasonality prior scale to 20.0, and the seasonality mode to multiplicative. Three features of the search are worth noting.

The changepoint prior scale dominates the outcome.
Every one of the five best-performing configurations sets it to the lowest value examined, indicating that a near-rigid trend suits these series better than a flexible one.

The seasonality prior scale makes almost no difference.
At the selected changepoint prior scale and mode, its three values differ by less than 0.003 in mean MASE. The size of the daily and weekly swings is therefore decided by the traffic data rather than by this setting.

The multiplicative mode is favourable only in combination with the lowest changepoint prior scale.
It is substantially worse than the additive mode at every other changepoint value. The selected configuration is the one point at which the two settings work together, and the margin it gains there is small. It was retained because the selection criterion was fixed before the search.

Table 4.10: Prophet at default and tuned configuration across the fifty cells

Measure	Default	Tuned
Cells with MASE < 1	29 / 50	33 / 50
Mean MASE	1.114	1.024
Median MASE	0.945	0.860

Table 4.10 reports the effect of the selected configuration on the held-out period. Five of the twelve cells that Section 4.4.4 identified as reached by another candidate but not by Prophet are recovered under the tuned configuration. One cell reached at the default configuration is no longer reached under the tuned one, so the net gain is four.

Tuning improves how accurately the model reproduces a cell's traffic. Whether a cell can be assessed by the detection rule is a separate question, and one that accuracy

alone does not answer. Two conditions are applied to each cell before deployment, both evaluated on the held-out period using the tuned model.

Accuracy.
The fitted model must improve on the naive baseline for that cell, that is, record a MASE below one.

Suitability.
The forecast must exceed the cell's activity threshold at least once across the held-out period.

The second condition is not implied by the first. A forecast can reproduce a cell's traffic closely overall and still remain below the activity threshold throughout. For such a cell, no onset can be judged at once, and no crossing exists to set a deadline, so every episode simply waits for the forecast's highest point. The rule then loses immediate detection entirely.

Cells that satisfy both conditions are assessed by the detection rule described in Section 4.4.3. For a cell that fails either condition, the forecast does not reliably show when the cell should be active, so it cannot be used to set a deadline. Such cells are assessed instead against a fixed period of persistent zero traffic, set by operator policy. Its value is an operational choice between faster reporting and fewer false alarms. Section 5.2 reports how the fifty cells divide between the two.

4.4.6 Pipeline Implementation

The preceding subsections established what the system decides. This one describes how it is implemented. The items required for detection are prepared once from recorded history, and the assessment itself runs as the reports arrive. Figure 4.7 shows the arrangement.

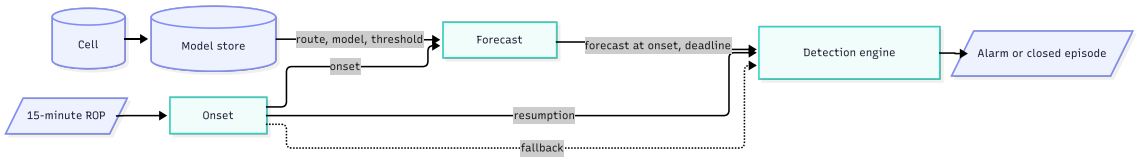


Figure 4.7: Arrangement of the detection system. The model store is written once for each cell from its recorded history and read thereafter, so no model fitting takes place while reports are being assessed. Cells routed to the fallback path bypass forecast generation.

Preparation begins with a cell's recorded history of the activity indicator. From this history the activity threshold is computed as described in Section 4.4.2, and the forecasting model is fitted with the configuration established in Section 4.4.5. The two conditions of Section 4.4.5 are then applied. A cell satisfying both is marked modellable, and its route, fitted model, and threshold are held for use during detection. A cell failing either is marked fallback, and only its route is held. This stage is completed before any incoming report is assessed.

Detection begins once preparation is complete. Reports arrive for each cell at fifteen-minute intervals. The value most recently seen for a cell is retained, so that an onset is recognised as a transition rather than as a property of a single report.

At an onset, the cell's route is consulted first. For a fallback cell, no forecast is generated, and the onset is assessed against the operator's fixed period.

For a modellable cell, a forecast covering the following 96 intervals is required. The cell may already have one from an earlier onset, which happens when traffic resumes briefly and then stops again before that onset's deadline. In that case the existing forecast and its deadline are reused, since the model has not changed and a new forecast would give the same values. Otherwise a forecast is generated from the onset, negative values are set to zero, and its deadline is set. The forecast value at the onset is then compared against the threshold, placing the episode on the immediate path where it exceeds the threshold and on the deferred path where it does not.

A timer tracks every open episode from its onset, and the episode closes as described in Section 4.4.3. A cell has one episode open at a time, and raising an alarm does not end observation of the cell.

A fault may also begin while the cell is already reporting zero. The system observes only what the counters report, so such a fault produces no transition and no new onset. It is still caught, because the whole zero period is treated as one episode. The episode opened at the earlier onset continues toward its deadline, and since the cell does not return to service, the deadline is reached and an alarm is raised. Table 4.11 sets out the situations in which a zero is observed at a modellable cell and how each resolves.

Each resolved episode is recorded with the context in which it was decided. This comprises the cell and its route, the onset timestamp, the forecast value at the onset and the cell's threshold, the deadline and which of the rules produced it, the path taken, the outcome, and where an alarm was raised, its timestamp and the detection delay.

Table 4.11: Situations in which a zero is observed and how each resolves

	Zero observed	Forecast consulted	Outcome
I	Expected active	Above threshold	Alarm at the onset
II	Expected quiet	At or below threshold	Alarm at the deadline, or none if traffic resumes
III	Within a normal zero	From the earlier onset	Resolved at that episode's deadline

Three boundaries apply to the method as described. Model fitting is carried out once from a cell's available history, and the behaviour of a fitted model as traffic patterns change over longer periods is not examined here. Because each cell is judged against its own level, a zero at one of its quieter hours is deferred even when the cell has considerable activity. The arrangement under which the system would be deployed across a network is an operational matter outside the scope of this work. This is a framework demonstrating the approach, not a deployment-ready implementation.

4.5 Demonstration and Evaluation (Activities 4 and 5)

The detection method described in Section 4.4 is put into operation on records drawn from the network. Two runs are carried out over the same week. In the first, faults are introduced at known times, so that the interval between a fault and the alarm reporting it can be measured. In the second, the records are left as captured, so that any alarm raised can be examined where no fault was introduced.

Running the system on these records is the demonstration. The quantities taken from the two runs are the evaluation. The objectives in Chapter 1 call for a fault to be reported within a bounded detection time, and for the approach to be assessed against the alarms it raises on cells that are legitimately idle. The first run supplies the first of these and the second run the other.

4.5.1 Evaluation Strategy

Section 4.2.4 established that the corpus carries no record of which cells were faulted. The system's decisions therefore cannot be checked against known outcomes, and detection performance cannot be measured directly from the records.

Measurement requires knowing when a fault occurred. Faults are therefore introduced at known times. Each is represented by setting the activity indicator to zero continuously from that moment. The timestamp is known, so the interval to the resulting alarm can be measured directly.

Every introduced fault is reported, because an alarm is raised whenever the indicator stays at zero until its deadline. This follows from how the method is built, not from the runs.

The first run measures how long reporting takes. A fault beginning when the cell is expected to be active is reported at once, and one beginning at any other time waits until the cell is next expected to be active, as described in Section 4.4.3. The delay therefore depends on where in the cell's cycle the fault began.

The forty-five days of records are divided into a fitting period, an evaluation window and a margin. A model representing a weekly cycle needs a history of several complete weeks, and the forty-five days available allow five weeks for fitting. Faults are introduced across a full week, since the system's response varies both across the hours of a day and across the days of a week. The three days remaining serve as a margin, so that an episode beginning at the last timestamp of that week can still reach its deadline, which Section 4.4.3 established as lying within one daily cycle. Table 4.12 sets out the division.

The evaluation window falls within the ten days held out from fitting in Sections 4.4.4 and 4.4.5. With forty-five days of records available and thirty-five required for fitting, no further period could be reserved for evaluation alone.

Both runs use the twenty-nine cells that Section 4.4.5 routed to forecast-based detection, each prepared as described in Section 4.4.6. The twenty-one cells routed to the fallback path are excluded. Their assessment rests on a waiting period set by operator policy, so a run on these cells would measure that policy rather than the method.

Table 4.12: Division of the observation period

Portion	Length	Purpose
Fitting	35 days	Five complete weeks, sufficient to estimate a weekly cycle
Evaluation window	7 days	One complete week, within which faults are introduced
Resolution margin	3 days	Allows episodes beginning at the end of the window to reach their deadline

4.5.2 Detection with Injected Faults

Each fault is introduced at a single timestamp, leaving the earlier records unchanged. A zero from that point onward is what a catatonic cell produces in the indicator, since it serves no users for the duration of the fault. No confirmed fault exists in the corpus to imitate, so the representation follows from this definition rather than from an observed case. The system then processes the modified series one report at a time, as it would in operation.

Every timestamp in the seven-day evaluation window serves as an injection point. This covers every position at which a fault could arise. A sample of injection points would leave the outcome dependent on which points were drawn. Since the window has a finite number of timestamps and the assessment is deterministic, every one can be tested, giving the outcomes themselves rather than estimates of them.

The injections are carried out separately for each of the twenty-nine cells, and each injection is a separate run. A cell's deadlines follow from its own forecast, so the delay a fault incurs is a property of that cell's activity pattern. Results are therefore reported per cell, so that differences between cells remain visible.

The quantity measured is the interval between the onset the system observed and the alarm it raised, which is zero when the alarm is raised at the onset. Some injection points fall inside a period the cell was already spending at zero. Nothing changes at that timestamp, so the system's onset is the earlier one, when the indicator first fell to zero. The interval is measured from that onset, the earliest point at which the system had anything to act on.

4.5.3 Detection on Unmodified Data

The same cells and the same week are processed with the records left as captured. The cells were provided as having operated normally throughout the observation period, as described in Section 4.4.1, so an alarm raised during this week is taken as an alarm raised where no fault was present.

The episodes examined are those present in the data, each beginning where the indicator falls from a positive value to zero. The run therefore covers the zero-traffic periods the system would encounter in operation.

A count of alarms establishes how often the system reported a fault where none

was present. A second quantity is taken from the episodes that closed without an alarm: the interval between traffic resuming and the deadline set for that episode. This shows how much of the waiting period remained unused when the cell returned to service, and so how near the system came to raising an alarm.

The runs show how the system responds to persistent faults of the kind described in Section 4.2.3.

Chapter 5

Results and Analysis

This chapter reports the outcomes of the analyses and runs described in Chapter 4. Section 5.1 reports the comparison of pre-zero windows, which contributes to RQ2. Sections 5.2 to 5.5 report the applicability of the detection method and the outcomes of the two runs described in Section 4.5, which together answer RQ3 and provide the evidence for RQ1. Both runs were carried out on the twenty-nine modellable cells over the same seven-day window.

Cell identifiers are anonymised. The four cells in Section 5.2 are labelled S1 to S4. The twenty-nine cells assessed by forecast are labelled M1 to M29, in ascending order of the longest delay recorded for each.

5.1 Behaviour Preceding a Zero-Traffic Onset

The comparison described in Section 4.3.5 was carried out on the 37,585 windows, separately for each detector.

The two detectors identify the same kinds of feature as most strongly separating. Features describing how erratic a counter is within the window, and how far it fell from its highest value to its last, occupy the highest ranks for both. Features describing the trend across the window, or the shift between its two halves, rank consistently lower. This agreement is obtained from methods that select candidates on different principles, which indicates the pattern reflects the structure of the data rather than the behaviour of either detector. The candidates therefore do share a consistent basis.

The differences are statistically significant. Mann–Whitney U tests give $p < 0.001$ for each of the ten highest-ranked features of both detectors. This establishes that the two groups are genuinely unalike, but it does not establish that they can be told apart. With 37,585 windows, a difference of almost any size will reach significance, since the test detects smaller and smaller differences as the sample grows [37]. Assigning an individual window to one group or the other requires the groups to occupy distinguishable regions of the feature space, which a difference between their averages does not by itself establish.

The strongest feature under Isolation Forest separates the two groups by 1.41 standard deviations, and under Local Outlier Factor by 0.59. The candidates are the ten per cent of windows each detector found most unusual. If fault-induced transitions formed a substantial group with a distinct pattern, these candidates would stand apart more sharply. At the separations found by the two detectors, the candidates and the remaining windows overlap considerably, so these features cannot be

used to classify a single window. The complete rankings for both detectors are given in Appendix A.2.

This finding is specific to the conditions under which it was obtained. It concerns two detection methods, 37 features, eight counters at fifteen-minute reporting granularity, and a window of four reports. Other signals, a finer granularity, or a longer window might reveal structure that these conditions cannot express. Features were also assessed individually, so it remains open whether some combination of them separates the groups where no single feature does. Within these conditions, the interval preceding a zero-traffic onset showed no difference sufficient to distinguish a fault-induced transition from a natural one.

5.2 Applicability

Before either run can be described, it is necessary to say which cells the method reaches. The two conditions of Section 4.4.5 divide the fifty cells, as reported in Table 5.1, into twenty-nine assessed by the detection rule and twenty-one assessed against a fixed waiting period.

Table 5.1: Routing outcome across the fifty cells

Outcome	Cells
Both conditions satisfied, assessed by forecast	29
Accuracy condition not satisfied	17
Accuracy satisfied, suitability not satisfied	4

Seventeen of the twenty-one fail the accuracy condition. None of them improves on the naive baseline under the tuned model. These cells use the operator-defined period and are not examined further.

The other four pass the accuracy condition but fail the suitability condition. Their forecasts never rise above the cell's activity threshold during the held-out period, so the detection rule finds no point at which the cell would ordinarily be clearly active. Table 5.2 gives the forecast maximum and the threshold for each.

Table 5.2: The four cells failing the suitability condition. The gap is the shortfall of the forecast maximum against the threshold, as a percentage of the threshold.

Cell	Forecast maximum	Threshold	Gap
S1	61.3	66.0	7%
S2	47.6	58.0	18%
S3	26.6	47.0	43%
S4	42.8	104.0	59%

The gap ranges from 7 per cent on S1 to 59 per cent on S4, and all four are routed to the fallback path. The runs reported below operate on the twenty-nine cells that satisfy both conditions.

5.3 Detection with Injected Faults

Every fifteen-minute timestamp in the seven-day window was used as an injection point on each of the twenty-nine cells, giving 19,488 runs. All of them resolved within the available data. As established in Section 4.5.1, every injected fault is reported, so the runs measure how long the reporting takes.

Across the 19,488 runs, covering the twenty-nine cells over the full week, alarms were raised immediately in 49.0 per cent of cases and at a later deadline in 51.0 per cent. The proportion varies widely between cells, from 10.0 per cent raised at the onset on M29 to 70.5 per cent on M1.

The variation reflects how long each cell's forecast stays above its threshold during the day. A cell that is busy for most of the day places most injection points in the immediate case, while a cell active for only a few hours places most of them in the deferred case. The split is therefore a property of each cell's traffic pattern, not of the method.

Each cell's longest delay ranges from 8 hours on M1 to 23 hours 45 minutes on M28 and M29. A cell's typical delay is considerably shorter than its worst case, with the mean deferred delay ranging from 3 hours 42 minutes to 10 hours 50 minutes across the twenty-nine cells. Table 5.3 gives the aggregate outcome, and Table B.1 gives the breakdown by cell.

Table 5.3: Aggregate outcome of the 19,488 injection runs across the twenty-nine cells

Quantity	Value
Injection runs	19,488
Alarms raised at the onset	49.0%
Alarms raised at a deadline	51.0%
Mean deferred delay, per cell	3h 42min to 10h 50min
Longest delay recorded	23h 45min

5.4 Detection on Unmodified Data

On the unmodified records, the cells operated normally throughout the week, as Section 4.5.3 sets out, so any alarm raised is a false alarm. The run produced 104 zero-traffic episodes. Of these, 101 closed when traffic resumed and three raised an alarm. All three came from one cell, M15, within a window of 2 hours 15 minutes on a single day. The other twenty-eight cells produced no alarm across the whole week. Table 5.4 gives the three onsets with the forecast value and the cell's threshold at each.

The forecast at each onset is above the threshold, so the rule placed all three on the immediate path. Figure 5.1 shows the observed series and the forecast for this cell over the surrounding period.

The observed traffic falls from its evening peak to zero over about three hours, while the forecast declines more gently into the overnight period and settles above

Table 5.4: The three onsets that raised an alarm on the unmodified week, all on cell M15. Onset times correspond to those marked in Figure 5.1.

Onset time	Forecast at onset	Threshold
01:45	85.1	63.0
02:45	76.4	63.0
04:00	70.8	63.0

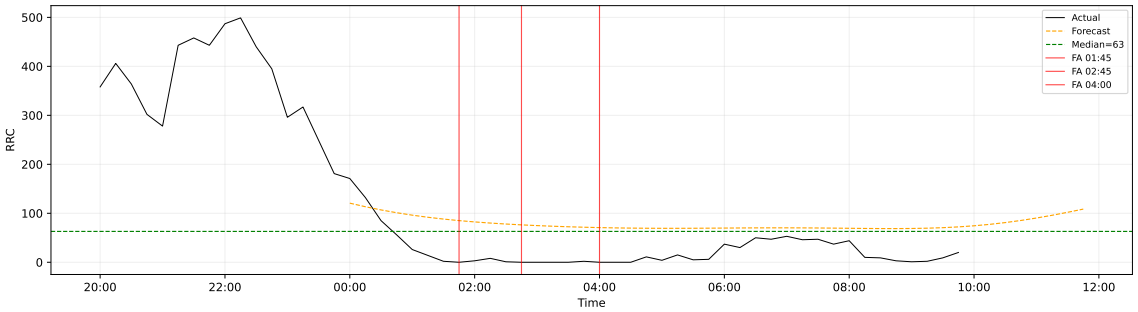


Figure 5.1: Cell M15 over the period containing the three alarms. The observed series and the forecast are shown against the cell's activity threshold (shown as the median), with the three false alarms (FA) marked.

the threshold. The gap between the two is what produced the alarms. The forecast still expected moderate activity through a period in which the cell reported none.

M15 satisfied both conditions of Section 4.4.5, both of which judge the forecast over the held-out period as a whole. Neither checks whether the forecast is right at any particular moment, and the three alarms came from the forecast being wrong at three specific onsets.

An episode on the deferred path stays open until its deadline and closes without an alarm if traffic resumes first. The immediate path gives no such opportunity, since the assessment is finished at the onset. In this run, the 101 episodes that closed without an alarm were all on the deferred path, and each closed before its deadline. The three alarms therefore came from the immediate path alone. A deferred episode can also produce a false alarm if a healthy cell stays silent beyond its deadline, and Section 5.5 reports how close the episodes in this run came to that point.

5.5 Margin on Resolved Episodes

The 101 episodes that closed without an alarm did so because traffic resumed before the deadline. The time remaining when it did is recorded as the margin, the interval between traffic resuming and the deadline set for that episode. A larger margin means the cell returned to service well before its deadline, and a smaller one that it returned shortly before. Margins are recorded at fifteen-minute granularity, since both resumption and deadline fall on report timestamps.

The smallest margin is 15 minutes and the median is 5 hours. Table 5.5 gives the distribution.

Table 5.5: Distribution of margin across the 101 episodes that closed without an alarm. Margins are counted in 15-minute reporting intervals.

Margin	Episodes
15 minutes	1
Above 15 minutes, up to 1 hour	0
Above 1 hour, up to 2 hours	7
Above 2 hours, up to 4 hours	32
Above 4 hours	61

Sixty-one of the 101 episodes cleared their deadline by more than four hours, and ninety-three by more than two. One episode fell in the smallest interval, on cell M5. The forecast at that onset was below the cell's threshold, so the episode went to the deferred path, and the first crossing above the threshold came 30 minutes after the onset and became the deadline. Traffic resumed 15 minutes before it. Figure 5.2 shows the episode.

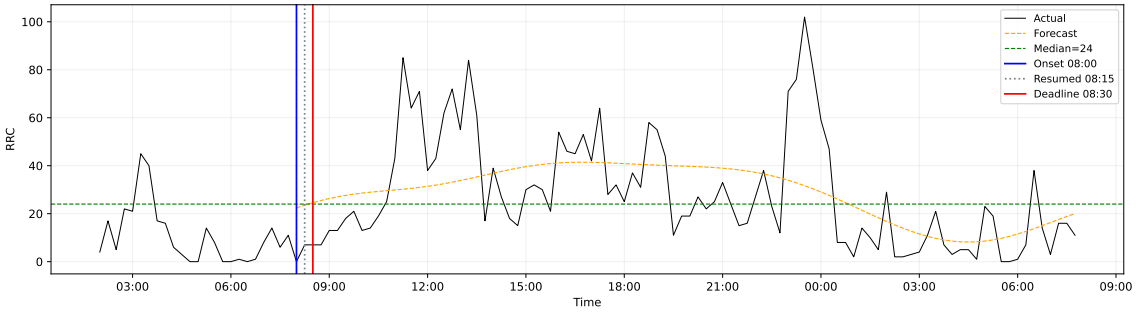


Figure 5.2: The episode with the smallest margin, on cell M5. The observed series and the forecast are shown against the cell's activity threshold, with the onset, the resumption and the deadline marked.

Taken together, the two runs describe how the method behaves on the twenty-nine cells it reaches. Across the injection runs, 49.0 per cent of alarms were raised at the onset and the rest at a deadline. The split on each cell followed the shape of its traffic, and mean deferred delays ranged from 3 hours 42 minutes to 10 hours 50 minutes across cells. On the same week left unmodified, three alarms came from one cell in a single window and twenty-eight cells produced none, while the episodes that closed without an alarm did so with a median margin of five hours.

These outcomes come from twenty-nine cells over one week of records, with faults introduced at known times rather than observed in the network. They establish how the method behaved under those conditions. The answers to the research questions are taken up in Chapter 6.

Chapter 6

Discussion

6.1 Answers to the Research Questions

The aim of this thesis was to enhance Advanced Cell Supervision through a method that uses each cell's own historical traffic pattern to judge whether an observed zero is expected or anomalous. The three research questions stated in Section 1.3 are answered below, each supported by the evidence produced in the preceding chapters.

RQ1: What AI-based approach can distinguish sleeping cells from healthy idle cells during zero-traffic periods in cellular networks?

An approach that learns each cell's own traffic pattern from its recorded history can distinguish the two conditions, because the pattern says what the cell would ordinarily be doing at the moment the zero occurs. The counters themselves are identical under both conditions and remain so. The difference lies in the moment. A cell falling silent when its own history says it should be carrying traffic is not in the same state as one falling silent when its history says it should be quiet. The distinction is not always available at the moment the zero arrives, and Section 6.2 examines what follows from that.

Making this operational requires two quantities for each cell, both derived from its own records. The first is the level of activity at which that cell counts as active, taken from what it reports when it is serving users. The second is the level it would ordinarily reach at any given moment, obtained from a model fitted to its history. An observed zero is then assessed by comparing the second against the first, and the same comparison determines how long to wait where the assessment cannot be made at once. Section 4.4 describes how each is derived.

The approach is learned rather than configured, in that both quantities come from the cell itself. For the cells it applies to, no operator sets a threshold or states when a cell ought to be busy, so each cell is judged against expectations learned from its own data. The approach applies to cells whose history yields a forecast that improves on repeating the previous day's values and that rises above the cell's own activity level at some point. Cells not meeting both conditions are assessed against a period the operator sets instead.

The supervision system in which this work is situated does not assess zero-traffic periods. The industrial partner identified this limitation and deferred it, and this work addresses it. Chapter 5 reports that across 19,488 injected faults on the twenty-nine cells to which the method applies, every fault was reported, as the construction of the rule ensures. On the same week left unmodified, 101 of 104 zero-traffic episodes

closed without an alarm. Taken together, the two runs show that the method separates the two conditions within the scope examined.

RQ2: What features or patterns extracted from cellular network data can effectively differentiate healthy zero-traffic cases from fault-related zero-activity conditions?

The two cases are differentiated by the timing of the zero relative to the cell's own established pattern, as set out in the answer to RQ1, and not by any feature of the reported values. Those features were examined first, in the interval preceding each onset.

Section 4.3 drew 37,585 intervals preceding a zero-traffic onset from 741 cells. Each was described through 37 features, and two detectors resting on different principles searched for structure among them. Section 5.1 reports the outcome. The candidates they selected shared a consistent basis, with both detectors ranking the same kinds of feature highest and Mann–Whitney tests confirming the differences as statistically significant. Significance, however, is not separation. At this sample size a difference of almost any size reaches significance, and the magnitude observed was 1.41 standard deviations on the strongest feature under one detector and 0.59 under the other, for the windows the detectors found most unusual. No feature examined identifies a subgroup of pre-zero intervals distinguishable from the rest. These features therefore offer no basis for telling a failed cell from an idle one before the zero begins.

This holds for the features and detectors used here, and for the eight counters at fifteen-minute granularity and the window of four reports. Features were assessed individually, so whether some combination separates the groups was not established. Within those conditions, the answer to RQ2 is that no feature of the reported values was found to differentiate the two cases, and the timing of the observation relative to the cell's pattern does.

RQ3: What detection performance does the proposed approach achieve in distinguishing healthy idle cells from sleeping cells during zero-traffic periods?

On the cells it reaches, the approach reports every persistent fault within one daily cycle, about half of them at the moment traffic stops, and raises false alarms rarely, with three across 104 zero-traffic episodes on the unmodified week, all on one cell.

The approach applies to cells whose forecast beats the naive baseline and rises above the cell's own activity level at some point. Twenty-nine of the fifty cells examined meet both conditions. The remaining twenty-one are assessed against a waiting period the operator sets instead. Because the conditions are checked before the method is used, an operator would know in advance which cells it covers.

On the twenty-nine cells, every introduced fault was reported, as the rule ensures, so the performance the runs measure is the delay. Roughly half the faults were reported at the onset and the rest at a deadline set by the cell's own forecast, depending on when in the cell's cycle the fault began. This share differs widely between cells, because their traffic patterns differ. Mean deferred delays ranged from under four

hours to nearly eleven, and no delay exceeded one daily cycle.

The three false alarms came from a single cell, within a window of just over two hours on one day, with the other twenty-eight producing none across the week. All three arose on the immediate path, on a cell that met both applicability conditions, where the forecast expected moderate activity through an early-morning period in which the cell reported none. The episodes that closed without an alarm did so with a median of five hours remaining before their deadline.

These outcomes were obtained with faults introduced at known times rather than observed, and on the same period used to select the model and the cells it applies to. Section 6.4 sets out what this bounds.

Taken together, on the cells it reaches, the approach replaces an unbounded wait with a bounded one, reporting most faults well within a day, at the cost of occasional false alarms where the forecast overestimates a quiet period.

6.2 Interpretation of the Detection Outcomes

The outcomes reported in Chapter 5 raise three questions about why the method behaves as it does, and each is taken in turn below.

A substantial share of the injected faults was not resolved at the onset, because at a moment the cell is ordinarily quiet nothing available settles the question. For these episodes, the contribution of the method lies in the point at which the wait ends. The deadline is the next time the cell is expected to be clearly active, which is the earliest moment at which continued silence departs from the cell's own established behaviour. Cells differ in when that moment arrives, so the wait differs between them, with mean deferred delays ranging from 3 hours 42 minutes to 10 hours 50 minutes across the twenty-nine cells. No fixed waiting period was evaluated in this work. Because the wait each cell needs differs, a single fixed period applied across the network would be expected to be too long for some cells and too short for others.

Some injected faults began while the cell was already reporting zero. Such a fault produces no transition, so the system has no new onset to act on. It is caught instead by the episode already open from the earlier onset and reported at that episode's deadline. The moment the fault began is not visible in the counters, so its delay can only be counted from the start of the silence.

All three false alarms came from the immediate path. That path raises its alarm at the onset, without waiting to see whether traffic returns, whereas the deferred path waits and closes without an alarm if traffic resumes, which is what all 101 deferred episodes did. A false alarm therefore arises when the forecast expects activity at a moment the cell is idle, as on M15 in the early hours. Three alarms in one week describe what was observed rather than establishing a rate. The activity threshold decides which path an onset takes, and its effect on false alarms was not examined here, as Section 6.5 notes.

6.3 Relation to Existing Work

Section 3.2 placed this work within the approaches that draw on the counters a cell reports about itself, since those remain available when a cell carries no traffic. With

the outcomes now reported, the work can be set against the approaches reviewed there.

The approaches drawing on device reports require devices present in the coverage area, since the measurements and events they use are produced by those devices. Fortes et al. record the consequence directly, reporting that narrowing the area they assess can leave too few device reports for a decision to be reached [13]. The approaches drawing on neighbouring cells require traffic to be displaced onto those neighbours and enough of it for the displacement to register, and Zhang, Feng et al. attribute their false negatives to cells carrying too little traffic for this [43]. No comparison of outcomes is available across these families, since the quantities each measures are not the same, but each carries a condition under which it applies and each records what follows where the condition is not met.

The counter-based approaches carry a condition of the same kind. Detection there rests on an expectation of what a cell ordinarily reports, against which a departure is identified, and Wu et al. [41] and Ming et al. [31] depend on that expectation being clearly above zero, as Table 3.1 records. At the hours a cell is ordinarily quiet, the expected value is itself close to zero. Hawulte reports that departures then go undetected [16], and Sangaiah et al. restrict their analysis to the busy hours for the same reason [35]. The method described here carries a condition of its own, that the cell's history supports a usable forecast. Twenty-nine of the fifty cells examined satisfy the two conditions of Section 4.4.5, and the remainder are assessed against a period the operator sets. The route each cell takes is decided before the method is applied. Section 6.5 notes that the twenty-one cells assessed against the operator's period were not evaluated, since their outcome depends on the period chosen.

Section 3.2 left two questions open within this family. The first was whether the interval leading up to a zero-traffic onset differs between a cell that has failed and one that has simply run out of traffic. This thesis examined it where none of the reviewed works do, across 37,585 such intervals. Under the conditions studied, it found no difference sufficient to tell the two apart, which narrows where within this family the information can lie. The second was whether a per-cell expectation can still be used once the observation is zero. In the framework of Le et al., the zero-traffic case is handled by a rule that classifies a cell as sleeping once its activity has stayed near zero for longer than a set duration, given as two hours by way of example. This rule overrides the framework's learned components, and its duration is set by operators rather than derived from the data [22]. This thesis derives the wait from each cell's own expectation. If a zero cannot be judged at once, the cell's forecast decides when the wait ends, so the wait differs from cell to cell, with mean deferred delays ranging from under four hours to nearly eleven.

Hawulte, the other work addressing this case, resolves the ambiguity by adding a measurement of uplink noise at the cell's own receiver, present even when no device is transmitting [16]. That route requires an additional measurement that the supervision data used in this thesis does not include. The two respond to the same condition and agree that traffic measurements alone cannot judge a quiet-hour zero at that moment. They differ only in the data each draws on.

Taken together, the reviewed work and the outcomes reported here point to the same constraint. At zero traffic, a silent cell cannot be judged from the moment alone, so any approach must draw on something beyond it. Hawulte adds further

evidence, and this thesis draws on the cell's own expected pattern. The device and neighbour families were evaluated on simulated networks. This thesis used records from an operational network but, as in the related works, introduced the faults. This is an interpretation drawn from the works reviewed in Chapter 3 and the runs reported in Chapter 5, not a general conclusion.

6.4 Threats to Validity

Five threats bear on the validity of the results. Each is stated below with its likely effect on the results. The limits of what the study covers are set out separately in Section 6.5.

Forecast quality measured over a period.
The accuracy condition judges a forecast by its average error across the held-out period. A forecast can be accurate on average and still wrong at the moment a zero-traffic onset occurs, and neither routing condition checks for this. Section 5.4 shows the effect. Cell M15 satisfied both conditions, yet its forecast expected activity at three onsets where the cell reported none, and each raised an alarm. Passing both conditions therefore does not rule out false alarms on a cell.

Delay measured from the observed onset.
A fault that begins while a cell is already reporting zero produces no observable transition. Its delay is therefore measured from the earlier onset, when the zero began. The recorded delay for such faults is accordingly longer than the time they actually went unreported.

Held-out days used for selection and evaluation.
Of the forty-five days of records, thirty-five were used for fitting and ten were held out. Those ten days were used both to choose the forecasting model (Section 4.4.4) and to decide which cells passed the two conditions of Section 4.4.5. The evaluation week also falls within them. The model and the twenty-nine cells were therefore chosen because their forecasts worked well on those held-out days, which were later also used for the evaluation. As a result, the reported outcomes are likely more favourable than they would be on unseen days. No further period could be set aside, since fitting a weekly cycle required thirty-five of the forty-five days.

Cells taken to be free of faults.
The fifty cells were identified by the industrial partner's domain experts as having operated normally, since the corpus records no faults. A fault in a cell's history would most likely make its forecast less accurate and the cell more likely to fail the accuracy condition. Any mistake in this identification would therefore work against the reported outcomes. Because the fifty were identified from a corpus of 1024, the twenty-nine reached may not reflect the corpus as a whole.

Few fault-induced windows among the candidates.
If fault-induced windows in the pre-zero analysis were very few relative to the 3,758 candidates each detector selected, they would rank among the candidates but could

go unnoticed in a comparison between the candidates as a whole and the rest. The absence of a separable group in Section 5.1 therefore does not exclude a pattern confined to a very small number of windows.

6.5 Limitations

The method operates on the performance counters a cell reports about its own operation, which is the data described in Section 4.2.2 and the extent of what was available to this work. The limitations below all lie within that boundary.

Data and evaluation period.
The study draws on forty-five days of counter records from a single LTE network, with the evaluation confined to one week, so how the method behaves on other networks, other radio technologies or over longer periods was not examined. The overlap between the held-out days and the evaluation, and the basis on which the fifty cells were taken to be free of faults, are set out in Section 6.4.

Synthetic fault injection.
Faults were introduced as a persistent zero from a known point, since the corpus contains no confirmed fault to use instead. The finding that every fault was reported applies to faults of this form. The method's response to real faults was not examined.

Fault conditions addressed.
The method addresses faults in which a cell reports zero from the onset onward. Intermittent faults, and faults in which a cell still serves some traffic, lie outside the scope of this thesis. Faults that affect users without changing the cell's counters are likewise beyond its reach.

Cells the method does not reach.
The method applies to twenty-nine of the fifty cells examined. The remaining twenty-one are assessed against a period set by the operator, and their behaviour was not evaluated, since a run on them would measure that setting rather than the method.

Model selection.
If each cell had its own best model, 41 of the fifty cells would outperform the naive baseline. Using one model class for all of them reduces this to 33, reached by the tuned model, and the other eight are assessed on the fallback path. The model was also selected on a single held-out period, so a different period might have led to a different choice. The detection rule itself does not depend on which model supplies the forecast.

Evidence on false alarms.
The alarms raised where no fault was present amount to three occurrences on one cell within a single week. That is the whole of the evidence on this point, and it describes what was observed rather than establishing a rate. All three arose on the immediate path, which the activity threshold governs. The threshold was held fixed, so how the count would change under a different definition remains to be examined.

Scope of the exploratory finding.
The finding reported in Section 5.1 holds for the two detectors applied, the thirty-seven features computed, the eight counters at fifteen-minute granularity and the window of four reports. Features were assessed individually, and their combinations, other signals, a finer granularity and a longer window were not examined.

6.6 Ethical, Societal, and Sustainability Aspects

Because the method works on aggregate cell-level counters, it needs no information about individual users to operate, and could be deployed without processing subscriber data. The data used in this work was handled as described in Section 1.4.1.

A failed cell that still appears operational leaves the users in its coverage area without service, including access to emergency services, and gives no indication of the fault. The longer it goes unreported, the greater the harm. On the cells it reaches, the method bounds that time, so detection no longer depends on a user complaint, and the cells it does not reach are identified before it is applied.

The sustainability impact described in Section 1.4.3 grows with the time a fault goes unnoticed, in both power and manual effort. Bounding that time would be expected to reduce it, although power and manual effort were not measured in this work.

Chapter 7

Conclusions and Future Work

This thesis investigated how a sleeping cell can be distinguished from a healthy idle cell during zero-traffic periods, a case that Advanced Cell Supervision does not currently assess. The work followed the design science research methodology in collaboration with Ericsson, using operational LTE counter data. It produced a detection method that judges each zero against the activity the cell's own forecast expects at that moment, and evaluated it with faults introduced at every timestamp of one week and on the same week left unmodified.

The interval before a zero-traffic onset did not indicate whether the cell had failed, under the conditions examined, so the distinction had to be drawn from elsewhere. The two cases are separated by the timing of the zero relative to the cell's own established pattern. A forecast fitted to that pattern puts this into practice. If a zero arrives when the cell would ordinarily be active, it is reported at once. If it arrives when the cell would ordinarily be quiet, it is held until the cell should have returned to service.

On the twenty-nine of the fifty cells the method reaches, every injected persistent fault was reported within one daily cycle, about half of them at once, as the rule ensures for faults of this form. On the unmodified week, three false alarms arose across 104 zero-traffic episodes, all on a single cell. The remaining twenty-one cells, whose history does not support a usable forecast, are identified before the method is applied and assessed against a waiting period set by operator policy.

7.1 Contributions

A negative result that narrows the problem.
The interval before a cell falls silent was examined with two detectors resting on different principles, and nothing was found there to separate a failed cell from an idle one. This narrows where the distinction can be found, and the work turned instead to temporal context as the basis for detection.

A detection method resting on temporal context.
The quantity that separates the two conditions is the activity a cell would ordinarily reach at the moment its traffic stopped, measured against the level at which that cell counts as active. Both are derived from the cell's own history, so for the cells the method reaches, no threshold is set by an operator and no waiting period is fixed in advance. When a zero cannot be judged at once, the wait ends at the earliest point

at which continued silence departs from the cell's own pattern, and always within one daily cycle.

An evaluation that yields outcomes rather than estimates.
With no confirmed fault in the corpus, faults were introduced at every timestamp of the evaluation window on every cell, and the same window was processed unmodified. Because the assessment is deterministic and every injection point was used, the runs give the outcome at each point, not an estimate from a sample.

An account of the blind spot.
The inability of supervision to separate the two conditions during zero traffic follows from what the counters record. They register the procedures a cell has handled, and a cell that has handled none produces the same record whether it had nothing to handle or was unable to handle anything. The blind spot is therefore a property of the measurement rather than a shortcoming in how it is read.

7.2 Practical Implications

For an operator, the method gives a decision for periods that supervision currently leaves unassessed. Zero traffic previously produced no signal at all. On the cells the method reaches, every zero-traffic episode now ends in a decision, either an alarm or a closure when traffic returns.

The timing of the decision follows when the silence begins. A fault that starts in a quiet period is reported if the cell stays silent past the time it should have resumed. For some cells, that is several hours after the fault began. This delay is the cost of not alarming on cells that are simply idle. The twenty-one cells the method does not reach remain with the operator's own waiting period. In deployment, the method therefore adds to that policy rather than replacing it.

The mechanism, judging a zero against the level a cell would ordinarily reach at that moment, draws on nothing specific to LTE and would be expected to apply wherever cells report counters that follow a regular daily pattern. The reported figures rest on one network, twenty-nine cells, one week and injected faults, and hold for those conditions.

7.3 Future Work

Extending the evidence.
Evaluating the method on a separate period of data, over several weeks, and on other networks and radio technologies would show how far the results extend. Testing it against confirmed faults, where such cases can be obtained, would show how it handles faults as they occur in operation.

Concept drift and refitting.
Establishing when a model no longer describes its cell, and how often refitting is required, would be needed for operation over extended periods.

Cells the method does not reach.
Twenty-one cells fall outside the method. Whether a different model class, horizon or activity indicator would serve them is open. Deriving their waiting period from the data, rather than setting it as policy, is also open.

Refining the assessment.
During a wait, the rule gives no signal until the deadline. As the forecast rises while the cell stays silent, the evidence of a fault grows, and this could be reported as a rising level of concern rather than a single alarm at the end, giving an operator a view of the case as it builds. The activity threshold that decides between immediate and deferred handling could also be tuned, and its effect on delays and false alarms examined.

Revisiting the exploratory finding.
Features in the pre-zero window were assessed individually. Whether some combination separates the two conditions, and whether other signals or granularities carry structure these conditions could not express, remains open.

References

[1] M. Amirijoo, L. Jorguseski, T. Kurner, R. Litjens, M. Neuland, L. C. Schmelz, and U. Turke, "Cell outage management in LTE networks," in *2009 6th International Symposium on Wireless Communication Systems*, 2009, pp. 600–604.

[2] A. Asghar, H. Farooq, and A. Imran, "Self-healing in emerging cellular networks: Review, challenges, and research directions," *IEEE Communications Surveys & Tutorials*, vol. 20, no. 3, pp. 1682–1709, 2018.

[3] M. Z. Asghar, M. Abbas, K. Zeeshan, P. Kotilainen, and T. Hämäläinen, "Assessment of deep learning methodology for self-organizing 5G networks," *Applied Sciences*, vol. 9, no. 15, p. 2975, 2019.

[4] G. Auer, V. Giannini, C. Desset, I. Godor, P. Skillermark, M. Olsson, M. A. Imran, D. Sabella, M. J. Gonzalez, O. Blume, and A. Fehske, "How much energy is needed to run a wireless network?" *IEEE Wireless Communications*, vol. 18, no. 5, pp. 40–49, 2011.

[5] B. Azkaei, K. C. Joshi, and G. Exarchakos, "Machine learning-driven anomaly detection for 5G O-RAN performance metrics," in *IEEE INFOCOM 2025 - IEEE Conference on Computer Communications Workshops (INFOCOM WKSHPS)*, 2025, pp. 1–6.

[6] K. Bandara, R. J. Hyndman, and C. Bergmeir, "MSTL: A seasonal-trend decomposition algorithm for time series with multiple seasonal patterns," *arXiv preprint arXiv:2107.13462*, 2021.

[7] R. Barco, P. Lazaro, and P. Munoz, "A unified framework for self-healing in wireless networks," *IEEE Communications Magazine*, vol. 50, no. 12, pp. 134–142, 2012.

[8] M. M. Breunig, H.-P. Kriegel, R. T. Ng, and J. Sander, "LOF: Identifying density-based local outliers," in *Proceedings of the 2000 ACM SIGMOD International Conference on Management of Data*, 2000, pp. 93–104.

[9] F. Chernogorov, S. Chernov, K. Brigatti, and T. Ristaniemi, "Sequence-based detection of sleeping cell failures in mobile networks," *Wireless Networks*, vol. 22, no. 6, pp. 2029–2048, 2016.

[10] S. Chernov, M. Cochez, and T. Ristaniemi, "Anomaly detection algorithms for the sleeping cell detection in LTE networks," in *2015 IEEE 81st Vehicular Technology Conference (VTC Spring)*, 2015, pp. 1–5.

[11] R. B. Cleveland, W. S. Cleveland, J. E. McRae, and I. Terpenning, "STL:

A seasonal-trend decomposition procedure based on loess," *Journal of Official Statistics*, vol. 6, no. 1, pp. 3–73, 1990.

[12] A. M. De Livera, R. J. Hyndman, and R. D. Snyder, "Forecasting time series with complex seasonal patterns using exponential smoothing," *Journal of the American Statistical Association*, vol. 106, no. 496, pp. 1513–1527, 2011.

[13] S. Fortes, R. Barco, and A. Aguilar-Garcia, "Location-based distributed sleeping cell detection and root cause analysis for 5G ultra-dense networks," *EURASIP Journal on Wireless Communications and Networking*, vol. 2016, no. 1, p. 149, 2016.

[14] S. Gregor and A. R. Hevner, "Positioning and presenting design science research for maximum impact," *MIS Quarterly*, vol. 37, no. 2, pp. 337–355, 2013.

[15] S. Hämäläinen, H. Sanneck, and C. Sartori, Eds., *LTE Self-Organising Networks (SON): Network Management Automation for Operational Efficiency.* Hoboken, N.J.: Wiley, 2011.

[16] Y. Hawulte, "Hybrid approach to detect fault caused KPIs anomaly in UMTS cells," M.S. Thesis, Addis Ababa Institute of Technology, Addis Ababa University, Addis Ababa, Ethiopia, 2020.

[17] A. R. Hevner, S. T. March, J. Park, and S. Ram, "Design science in information systems research," *MIS Quarterly*, vol. 28, no. 1, pp. 75–106, 2004.

[18] R. J. Hyndman and G. Athanasopoulos, *Forecasting: Principles and Practice*, 2nd ed. Melbourne, Australia: OTexts, 2018. [Online]. Available: https://OTexts.com/fpp2/

[19] ——, *Forecasting: Principles and Practice*, 3rd ed. Melbourne, Australia: OTexts, 2021. [Online]. Available: https://otexts.com/fpp3/

[20] R. J. Hyndman and A. B. Koehler, "Another look at measures of forecast accuracy," *International Journal of Forecasting*, vol. 22, no. 4, pp. 679–688, 2006.

[21] B. Kitchenham and S. Charters, "Guidelines for performing systematic literature reviews in software engineering," Keele University and Durham University, EBSE Technical Report EBSE-2007-01, 2007.

[22] N. Q. Le, D. T. Truong, T. P. Dinh, A. T. Nguyen, and V. T. Tran, "A hybrid AI-based framework for robust sleeping cell detection in 5G networks," Research Square preprint, 2025.

[23] C. Leys, C. Ley, O. Klein, P. Bernard, and L. Licata, "Detecting outliers: Do not use standard deviation around the mean, use absolute deviation around the median," *Journal of Experimental Social Psychology*, vol. 49, no. 4, pp. 764–766, 2013.

[24] F. T. Liu, K. M. Ting, and Z.-H. Zhou, "Isolation forest," in *Proceedings of the 2008 Eighth IEEE International Conference on Data Mining.* IEEE, 2008, pp. 413–422.

[25] A. A. Mahajan, "Hybrid model using LOF and iForest algorithms for detection of insider threats," TechRxiv preprint, 2024.

[26] S. Makridakis, E. Spiliotis, and V. Assimakopoulos, "The M4 competition: 100,000 time series and 61 forecasting methods," *International Journal of Forecasting*, vol. 36, no. 1, pp. 54–74, 2020.

[27] M. Malekzadeh and A. A. Abdul Ghani, "3-sector cell vs. omnicell: Cell sectorization impact on the performance of side-by-side unlicensed LTE and 802.11ac air interfaces," *IEEE Access*, vol. 7, pp. 122 315–122 329, 2019.

[28] O. G. Manzanilla-Salazar, F. Malandra, H. Mellah, C. Wetté, and B. Sansò, "A machine learning framework for sleeping cell detection in a smart-city IoT telecommunications infrastructure," *IEEE Access*, vol. 8, pp. 61 213–61 225, 2020.

[29] U. Masood, A. Asghar, A. Imran, and A. N. Mian, "Deep learning based detection of sleeping cells in next generation cellular networks," in *2018 IEEE Global Communications Conference (GLOBECOM)*, 2018, pp. 206–212.

[30] Meta Open Source, "Prophet: Forecasting at scale — documentation," https://facebook.github.io/prophet/docs/, 2026, accessed: 2026-08-27.

[31] Z. Ming, C. Sun, X. Li, Q. Fan, X. Wang, and V. C. M. Leung, "Ensemble learning based sleeping cell detection in cloud radio access networks," in *2020 IEEE Symposium on Computers and Communications (ISCC)*, 2020, pp. 1–6.

[32] J. Moysen and L. Giupponi, "From 4G to 5G: Self-organized network management meets machine learning," *Computer Communications*, vol. 129, pp. 248–268, 2018.

[33] K. Peffers, T. Tuunanen, M. A. Rothenberger, and S. Chatterjee, "A design science research methodology for information systems research," *Journal of Management Information Systems*, vol. 24, no. 3, pp. 45–77, 2007.

[34] P. Runeson and M. Höst, "Guidelines for conducting and reporting case study research in software engineering," *Empirical Software Engineering*, vol. 14, no. 2, pp. 131–164, 2009.

[35] A. K. Sangaiah, S. Rezaei, A. Javadpour, F. Miri, W. Zhang, and D. Wang, "Automatic fault detection and diagnosis in cellular networks and beyond 5G: Intelligent network management," *Algorithms*, vol. 15, no. 11, p. 432, 2022.

[36] S. Sesia, I. Toufik, and M. Baker, *LTE – The UMTS Long Term Evolution: From Theory to Practice*, 2nd ed. John Wiley & Sons, 2011.

[37] G. M. Sullivan and R. Feinn, "Using effect size—or why the P value is not enough," *Journal of Graduate Medical Education*, vol. 4, no. 3, pp. 279–282, 2012.

[38] S. J. Taylor and B. Letham, "Forecasting at scale," *The American Statistician*, vol. 72, no. 1, pp. 37–45, 2018.

[39] X. Wang, K. Smith, and R. Hyndman, "Characteristic-based clustering for time series data," *Data Mining and Knowledge Discovery*, vol. 13, no. 3, pp. 335–364, 2006.

[40] C. Wohlin, P. Runeson, M. Höst, M. C. Ohlsson, B. Regnell, and A. Wesslén, *Experimentation in Software Engineering*. Springer, 2012.

[41] J. Wu, P. P. C. Lee, Q. Li, L. Pan, and J. Zhang, "CellPAD: Detecting performance anomalies in cellular networks via regression analysis," in *2018 IFIP Networking Conference (IFIP Networking) and Workshops*, 2018, pp. 1–9.

[42] F. Xu, Y. Li, H. Wang, P. Zhang, and D. Jin, "Understanding mobile traffic patterns of large scale cellular towers in urban environment," *IEEE/ACM Transactions on Networking*, vol. 25, no. 2, pp. 1147–1161, 2017.

[43] T. Zhang, L. Feng, P. Yu, S. Guo, W. Li, and X. Qiu, "A handover statistics based approach for cell outage detection in self-organized heterogeneous networks," in *Proceedings of the 2017 IFIP/IEEE Symposium on Integrated Network and Service Management (IM)*. IEEE, 2017, pp. 628–631.

[44] T. Zhang, K. Zhu, and D. Niyato, "Detection of sleeping cells in self-organizing cellular networks: An adversarial auto-encoder method," *IEEE Transactions on Cognitive Communications and Networking*, vol. 7, no. 3, pp. 739–751, 2021.

Appendix A

Supplemental Information

A.1 Pre-Zero-Traffic Feature List

This section lists the 37 features computed for the pre-zero window analysis described in Section 4.3. Identifiers are those used in the implementation.

Each of the 32 temporal features is identified by a counter prefix and a measure suffix joined by an underscore, so that `RACH_Att_vol` denotes the volatility of RACH STP Attempts. Every combination of prefix and suffix exists. The prefixes are given in Table A.1 and the suffixes in Table A.2. The four measures are described in Table 4.4.

Table A.1: Counter prefixes used in feature identifiers

Counter	Prefix
RACH STP Attempts	`RACH_Att`
RACH STP Fails	`RACH_Fai`
RRC Conn Setup Attempts Total	`RRC_Att`
RRC Conn Setup Successes	`RRC_Succ`
Call Attempts	`Call_Att`
DL Traffic Volume	`DL_Vol`
RACH CBRA Success Rate	`CBRA_Rate`
RRC-to-RACH Attempt Ratio	`RRC_RACH`

Table A.2: Measure suffixes used in feature identifiers

Measure	Suffix
Slope	`slope`
Volatility	`vol`
Peak-to-final drop	`maxfinal`
Half-to-half difference	`halfdiff`

The five remaining features do not follow this pattern and are listed individually in Table A.3. Cross-signal features are computed from raw counter values at the final report of the window, with each denominator floored at one. Divergence features are computed from the normalised slopes.

Table A.3: Cross-signal and divergence features

Group	Identifier	Definition
Cross-signal	cross_rach_fail_ratio	RACH_Fai / RACH_Att
Cross-signal	cross_rrc_succ_ratio	RRC_Succ / RRC_Att
Cross-signal	cross_cbra_final	CBRA_Rate
Divergence	div_norm_rach_att_minus_cbra	RACH_Att_slope − CBRA_Rate_slope
Divergence	div_norm_rrc_att_minus_succ	RRC_Att_slope − RRC_Succ_slope

A.2 Feature Rankings

This section reports the feature rankings from the anomaly detection described in Section 4.3.4. The strongest scaled differences are stated in Section 5.1.

Tables A.4 and A.5 give the ranking of all 37 features for each detector at the 10% threshold. Values are the mean difference between the candidate group and the remaining windows, expressed in standard deviations on the standardised feature matrix. Each ranking is computed on that detector's own candidate set.

Table A.4: Feature rankings for Isolation Forest at the 10% threshold

Rank	Feature	Difference
1	RACH_Att_vol	1.4060
2	RACH_Fai_vol	1.3404
3	RRC_Succ_vol	1.2382
4	RRC_Att_vol	1.2346
5	RRC_Succ_maxfinal	1.1220
6	RRC_Att_maxfinal	1.1175
7	RACH_Att_maxfinal	1.0968
8	Call_Att_vol	1.0575
9	RACH_Fai_maxfinal	1.0103
10	Call_Att_maxfinal	0.9969
11	CBRA_Rate_vol	0.9366
12	CBRA_Rate_maxfinal	0.8882
13	RRC_RACH_vol	0.7257
14	DL_Vol_maxfinal	0.7246
15	DL_Vol_vol	0.7241
16	RRC_Succ_slope	0.6605
17	RRC_Att_slope	0.6544
18	Call_Att_slope	0.6210
19	RRC_RACH_maxfinal	0.5841
20	RRC_Succ_halfdiff	0.5491
21	RRC_Att_halfdiff	0.5428
22	Call_Att_halfdiff	0.5183
23	CBRA_Rate_slope	0.4524
24	div_norm_rach_att_minus_cbra	0.4194

Rank	Feature	Difference
25	CBRA_Rate_halfdiff	0.3566
26	cross_rach_fail_ratio	0.3510
27	RACH_Att_slope	0.3434
28	RACH_Fai_slope	0.2807
29	RACH_Att_halfdiff	0.2713
30	cross_cbra_final	0.2627
31	RACH_Fai_halfdiff	0.2137
32	DL_Vol_slope	0.1860
33	div_norm_rrc_att_minus_succ	0.1643
34	DL_Vol_halfdiff	0.1557
35	cross_rrc_succ_ratio	0.0976
36	RRC_RACH_slope	0.0759
37	RRC_RACH_halfdiff	0.0713

Table A.5: Feature rankings for Local Outlier Factor at the 10% threshold

Rank	Feature	Difference
1	RACH_Fai_vol	0.5900
2	DL_Vol_vol	0.5884
3	RACH_Att_vol	0.5874
4	CBRA_Rate_vol	0.5399
5	RRC_Succ_vol	0.4974
6	RRC_RACH_vol	0.4909
7	DL_Vol_maxfinal	0.4883
8	RRC_Att_vol	0.4873
9	RRC_RACH_maxfinal	0.4137
10	Call_Att_vol	0.3662
11	CBRA_Rate_maxfinal	0.3412
12	RRC_Succ_maxfinal	0.3206
13	RRC_Att_maxfinal	0.3146
14	RACH_Att_maxfinal	0.2844
15	RACH_Fai_maxfinal	0.2668
16	Call_Att_maxfinal	0.2531
17	cross_rach_fail_ratio	0.1641
18	CBRA_Rate_halfdiff	0.1336
19	cross_rrc_succ_ratio	0.1271
20	CBRA_Rate_slope	0.1221
21	div_norm_rach_att_minus_cbra	0.1210
22	RRC_Succ_slope	0.1018
23	RRC_Att_slope	0.0984
24	Call_Att_slope	0.0928
25	RRC_Succ_halfdiff	0.0725
26	RRC_Att_halfdiff	0.0707

Rank	Feature	Difference
27	Call_Att_halfdiff	0.0678
28	RRC_RACH_slope	0.0620
29	div_norm_rrc_att_minus_succ	0.0532
30	RRC_RACH_halfdiff	0.0527
31	cross_cbra_final	0.0508
32	DL_Vol_slope	0.0439
33	DL_Vol_halfdiff	0.0432
34	RACH_Fai_slope	0.0315
35	RACH_Fai_halfdiff	0.0140
36	RACH_Att_halfdiff	0.0058
37	RACH_Att_slope	0.0051

A.3 Model Comparison Breakdown

Table A.6 shows which candidates reach each cell, supporting the coverage figures reported in Section 4.4.4. A cell is reached by a candidate when that candidate achieves MASE below one for it.

Table A.6: Cells reached, by combination of candidates

Candidates reaching the cell	Cells
All three	8
MSTL+ETS and Prophet	8
TBATS and Prophet	7
MSTL+ETS and TBATS	0
Prophet only	6
TBATS only	7
MSTL+ETS only	5
None	9

Appendix B

Per-Cell Injection Results

Table B.1 reports the per-cell breakdown of the 19,488 injection runs described in Section 5.3. Each row corresponds to one modellable cell. Cells are labelled M1 to M29 in ascending order of max delay, with ties broken by ascending mean deferred delay.

Table B.1: Per-cell breakdown of the injection runs across the 29 modellable cells. Immediate is the share of injection runs on the cell whose onset was on the immediate path. Deferred is the complement.

Cell	Immediate	Deferred	Max delay	Mean deferred delay
M1	70.5%	29.5%	8h 00min	3h 42min
M2	53.3%	46.7%	9h 45min	4h 04min
M3	63.1%	36.9%	9h 45min	4h 34min
M4	69.0%	31.0%	10h 00min	4h 23min
M5	62.8%	37.2%	10h 30min	4h 39min
M6	57.9%	42.1%	10h 45min	5h 11min
M7	58.5%	41.5%	11h 15min	5h 08min
M8	56.1%	43.9%	11h 15min	5h 24min
M9	57.9%	42.1%	12h 00min	5h 15min
M10	53.6%	46.4%	12h 00min	5h 43min
M11	53.7%	46.3%	12h 00min	5h 43min
M12	50.3%	49.7%	12h 15min	6h 06min
M13	51.2%	48.8%	12h 30min	6h 01min
M14	48.8%	51.2%	13h 00min	6h 19min
M15	56.1%	43.9%	14h 15min	7h 16min
M16	49.6%	50.4%	14h 30min	6h 17min
M17	45.7%	54.3%	14h 30min	6h 44min
M18	57.3%	42.7%	15h 00min	5h 47min
M19	38.8%	61.2%	15h 15min	7h 29min
M20	43.5%	56.5%	16h 00min	8h 13min
M21	38.4%	61.6%	16h 15min	7h 36min
M22	57.7%	42.3%	17h 15min	5h 52min
M23	50.6%	49.4%	18h 30min	6h 27min
M24	44.6%	55.4%	18h 30min	7h 09min
M25	27.5%	72.5%	21h 30min	8h 18min
M26	32.9%	67.1%	21h 30min	8h 29min

Cell	Immediate	Deferred	Max delay	Mean deferred delay
M27	31.5%	68.5%	22h 15min	7h 29min
M28	29.2%	70.8%	23h 45min	8h 59min
M29	10.0%	90.0%	23h 45min	10h 50min

Appendix C

Prompt Examples

The following examples illustrate how the Claude Opus 5 model was used. Responses are shown in shortened form.

C.1 Example: Rephrasing for Academic Tone

Prompt:

> Hey, review the following text from my thesis report and help me correct grammatical mistakes, awkward clauses or phrasing. Provide a refinement and also a suggested alternative to it. Make sure you follow understandable academic English standards without changing the meaning or missing any piece of information.
>
> "Each detector divides the windows by construction, selecting a fixed proportion of candidates even if they don't have any meaningful division between them. The presence of candidates themselves isn't informative itself. The analysis examines if the selected windows share a consistent division for having been selected, and if that division basis is significantly enough to classify each individual window."

Response:

> Four issues found:
>
> 1. Informal tone: the contractions "don't" and "isn't" should be written in full
> 2. Redundancy: "themselves" and "itself" repeat the same idea
> 3. "if" should be "whether" when introducing alternatives
> 4. Unclear phrasing: "division basis is significantly enough" is ungrammatical, and "between them" has no clear reference

Refinement: "Each detector divides the windows by construction, selecting a fixed proportion of candidates whether or not any meaningful division exists. The presence of candidates is therefore not informative in itself. The analysis examines whether the selected windows share a consistent basis for having been selected, and whether that basis is pronounced enough to classify each individual window."

Alternative: "By construction, each detector selects a fixed proportion of the windows as candidates, even when no meaningful division exists. The presence of candidates alone therefore carries no information. The analysis instead examines whether the selected windows were selected for a consistent reason, and whether that reason is strong enough to classify each window individually."

C.2 Example: Proofreading for Formatting and Punctuation

Prompt:

Help me proofread the text from my thesis report. Please identify any punctuation mistakes, spelling mistakes, inconsistencies between British and American spelling (I am writing in British), missing or double spaces, and incorrect capitalisation. Give the identified issues clearly, line by line, with the fix.

"the runs measure how long it takes to report. Across the 19488 runs alarms were raised immediately on 49 percent of cases and at a later deadline in 51.0 per cent. The proportion varies widely between cells from 10.0 percent at onset on M29 to 70.5 percent on M1"

Response:

Five issues found:

1. Capitalisation and full stop: "the runs" should be "The runs", and a full stop is missing after "M1"
2. British spelling: "percent" should be "per cent" (three times)
3. Numbers: "19488" should be "19,488", and "49" should be "49.0" to match "51.0"
4. Missing commas: "runs alarms" should be "runs, alarms", and "cells from" should be "cells, from"

5. Wording: "on 49 percent" should be "in 49.0 per cent", and "at onset" should be "raised at the onset"

Corrected version: "The runs measure how long it takes to report. Across the 19,488 runs, alarms were raised immediately in 49.0 per cent of cases and at a later deadline in 51.0 per cent. The proportion varies widely between cells, from 10.0 per cent raised at the onset on M29 to 70.5 per cent on M1."



Faculty of Computing, Blekinge Institute of Technology, 371 79 Karlskrona, Sweden